



Aid, Water, and Peace: Assessing Foreign Assistance's Impact on Conflict in Water-Stressed India - A Spatial Grid Analysis

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Abstract

Using the grid analysis with fixed effects method, this paper investigates the effect of foreign aid project allocation on conflict onset in the context of varying water stress situations in India. The study finds that allocating aid projects in areas without water stress (the base category, level 0) heightens the probability of conflict incidents by 2.983 percentage points. Nevertheless, if the projects are well-targeted within low-to-medium water stress areas (level 1), the net marginal effect reverses to -2.899 percentage points, outweighing the baseline conflict-inducing tendency. In extreme water stress areas (level 4), the net effect is near zero ($+0.039$ percentage points), suggesting that aid nearly fully offsets its baseline conflict-inducing effect but falls short of achieving a clear pacifying outcome. We also tested this method in India's neighboring countries in the South Asia region. The heterogeneous patterns of the results for each country demonstrate that one has to take into account the particular situation of each country when interpreting the diverse result patterns. For broader context, under the trend of revising the foreign aid policy of donor countries to be more aligned with their own interests, recipient countries should consolidate their efforts on developing their economy and tackling their social issues while leveraging slowly decreasing external support to achieve sustainable progress and stability.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Background	4
1.2.1	Water stress	4
1.2.2	Conflict	5
1.2.3	Foreign aid as a solution	6
1.2.4	Literature review and contribution of the paper	6
2	Methodology and Data	7
2.1	Data	7
2.1.1	Conflict	7
2.1.2	World Bank aid projects	9
2.1.3	Water stress	12
2.1.4	Temperature	12
2.1.5	Precipitation	13
2.1.6	Cropland percentage	13
2.1.7	GDP	14
2.1.8	Population	14
2.1.9	Descriptive Statistics	15
2.2	Methodology	15
3	Analytical results	19
3.1	Main results	19
3.1.1	Conflict Dummy	19
3.1.2	Other auxiliary dependent variables	22
3.2	Robustness tests	25
3.3	External validity	28
4	Conclusion	31
5	Reference	32
6	Appendix	36
6.1	External validity - 25 by 25 km grid cells	36
6.2	Maps and graphs of other South Asian countries	37

1 Introduction

1.1 Motivation

The aim of this study is to investigate the effect of foreign aid allocation on the onset of potential conflict, especially in the context of water stress in India, contributing to the literature by providing new insights into the spatial dynamics of aid and conflict under varying water stress conditions. This study uses the 50 by 50 km grid cell as the default unit of identification. Control variables such as GDP, population, representing economic growth and urbanization, and cropland percentage, representing the agriculture sector, and average temperature and monthly precipitation, representing natural conditions are included. The linear probability model was chosen for its simplicity in interpreting coefficients. In the main model, conflict is described by a dummy variable at the grid cell level, indicating whether conflict occurs (100) or not (0). The original dataset also provides a further breakdown into three sub-categories of conflict and measures conflict intensity via casualties, which are also tested as dependent variables in this study. We observed the aid data on the annual basis from 1995 to 2014, based on the availability of the Aid Data [2017](#), allowing for control of fixed-effect at the grid cell- and regional-time trend levels. Standard error was being clustered at the grid cell level to rule out heterogeneity within each cell. Besides the main results of India, we also applied the same regressions to neighboring countries in the South Asia region to test the external validity of the previous findings. Additionally, robustness checks were done by running the regressions with smaller grid cell resolution.

The main result shows that an allocation of a foreign aid project in a grid cell with no water stress (the base category, level 0) accelerates the potential of conflict onset by 2.983 percentage points. However, this effect is substantially modified when projects are placed within areas of different water stress levels. Specifically, when interaction terms with water stress categories are taken into account, projects located in areas with low-to-medium water stress (level 1) reduce the net probability of conflict by 2.899 percentage points (i.e. $2.983 - 5.882 = -2.899$), and projects in extreme water stress areas (level 4) reduce it by approximately 0.039 percentage points net. This means that well-targeted foreign aid in water-constrained areas can mitigate rather than exacerbate conflict. Among the control variables, the statistical significance of the variable log of GDP showcases that a 1 % percent increase in economic growth translates to a decrease in the probability of having conflict by 0.3881 %. Overall, the main result gives us a clear picture for the foreign aid administrator about the importance of identifying resource-constrained areas when distributing aid resources. To interpret the result, aid projects in low-to medium-water-stressed areas could be a tool, providing local communities with knowledge, training, and infrastructure as preventive measures in the future. On the other hand, projects in areas with extreme water stress could be a target for the opposition for financial seizure. Furthermore, it is suggested from the main result that effective governance and facilitation for economic growth from the local government reduce the likelihood of insurgency. Thus, alongside external aid, fostering local economic development presents itself as an organic strategy to lower conflict over time, offering a dual approach to peace-building in India.

1.2 Background

1.2.1 Water stress

Water stress, being defined as when the demand for water surpasses the available amount of water during a certain period of time, could lead to the deterioration of both the quantity and quality of water resources (WRI 2025). This phenomenon is widespread in many countries around the globe, with at least 50% of the global population (4 billion people) enduring the highly water-stressed conditions for at least one month in a year; 25% of the world experiences extremely high water stress situations, whereby they use over 80% of their annual renewable freshwater supply (UNESCO World Water Assessment Programme 2024). These numbers are not uniformly distributed: water stress is particularly acute in South and Central Asia, the Middle East and North Africa, and sub-Saharan Africa, regions that together account for a disproportionate share of the world's water-insecure population (Kuzma et al. 2023). This phenomenon has been driven by both exacerbating natural events and the intensification of human activities. For instance, droughts are becoming increasingly common, intense, and widespread, with the geographic concentration of impacts falling heavily on already water-scarce regions in the global south. According to Zaveri, Damania, and Engle 2023, over the last 50 years, extreme dry rainfall shocks have increased by 233% on average. Similarly to droughts, it is also reported with high confidence that heavy precipitation events (floods) have become more frequent and intense over most land areas since the 1950s. Furthermore, climate change is only about to intensify the global water cycle, thus increasing the intensity and frequency of extreme hydro-meteorological events like droughts and floods, especially in already water-stressed regions (Seneviratne et al. 2021, p.19).

Under these circumstances stands the ever-intensifying human activities. Overall, the demand for water consists mostly of agriculture, which accounts for 70 % of freshwater withdrawals, around 20 % for industrial applications, and 10% for domestic usage. Particularly, the agriculture sector is susceptible to water risks, where dependence on rainfed crop could be severely punished during times of drought, and bad water governance on water usage for irrigation systems could undermine the food security for many countries. The latter two sectors are primarily increasing the demand for water because of the rapid industrialization and urbanization, hence the expansion of the water supply and sanitation system in many parts of the developing world. Depending on the assumed scenarios, the global water demand is forecasted to be heightened by between 20% and 33% by 2050 compared to 2010 (Burek et al. 2016, p.69). Water stress, as previously described, poses several challenges to be overcome. Firstly, the trans-boundary nature of water resources—where countries or regions share river basins or aquifers—can lead to social, economic, and political disputes over water rights, allocation, and management, occurring at either an inter-regional or international level. Secondly, the disparity in allocation of water access between urban and rural areas could create the division and tension within one country or region, whereby demand for agricultural irrigation in the rural area could be subordinated to the domestic and industrial water usage for urban areas. Likewise, water stress could also drive migration from water-scarce regions to more abundant resource ones, further stressing the infrastructure of the host regions, which could lead to social unrest and division between the newcomers and the host communities (UNESCO World Water Assessment Programme 2024, p.20). Finally, the convergence of water scarcity and climate change will only make water governance a more difficult task.

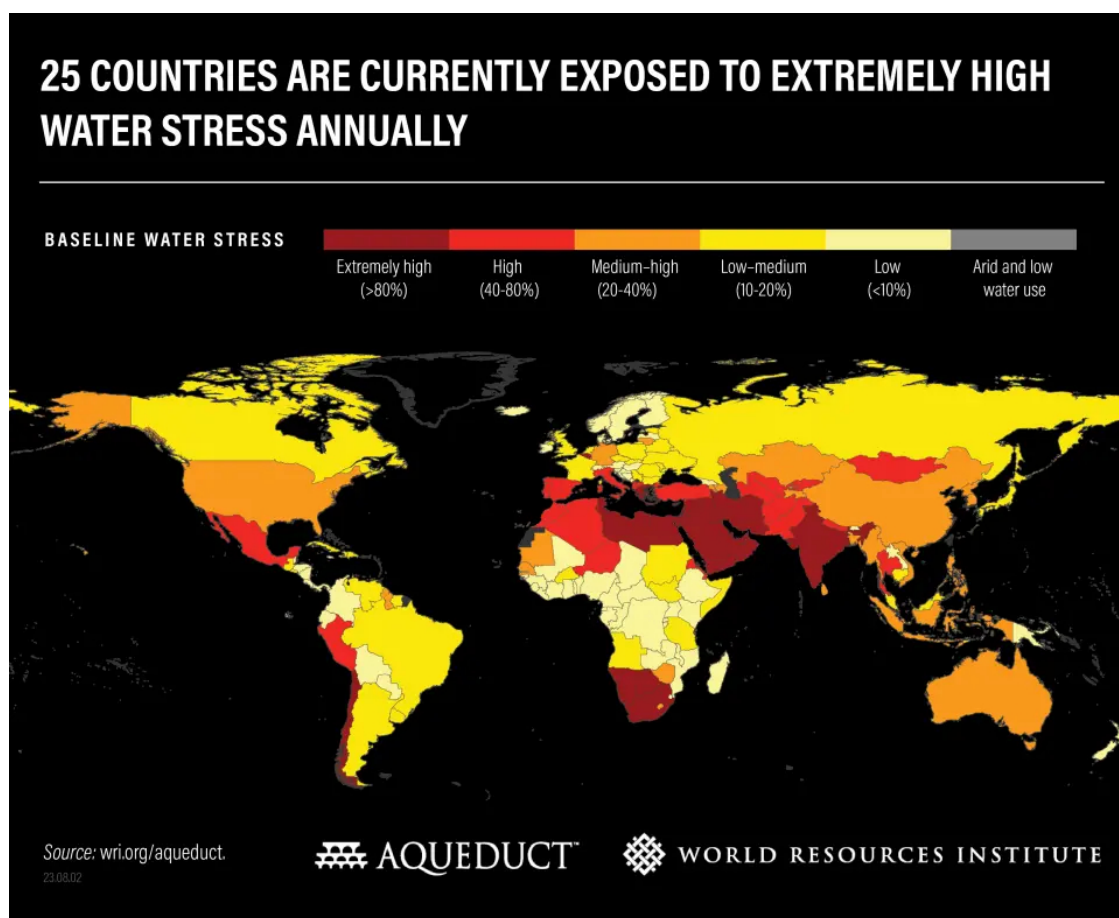


Figure 1: Visualization of water stress situation around the world

1.2.2 Conflict

Conflicts can take the form of interpersonal disagreements to outright large-scale wars due to different reasons, from resource competition, political and ideological differences, racial and ethnic discrimination, or socio-economic disparities. Given the introduction to the problem of water stress in the first paragraph, it becomes possible that it could cause disputes or conflicts among the population over resource allocation and management. For instance, the competition for water resources in the Nile Basin has historically contributed to tensions between Egypt, Sudan, and Ethiopia, making the negotiations for water management between three countries challenging and sometimes leading to threats of conflicts (UN 2021). According to the report of UNESCO World Water Assessment Programme 2024, water scarcity often co-moves with other factors, which impacts the onset of potential conflicts, so the direct causal effect of water stress on conflicts is difficult to affirm. The link between water supply and conflict has also been documented at the project level: Duflo and Pande (2007) show that large dam construction in India, by redistributing water access between upstream and downstream districts, generated significant political conflict and social tensions, underscoring how interventions in water infrastructure can have unintended conflict consequences. Likewise, regions with ongoing conflicts often lack a strong governance capability among the public institutions; essential infrastructure for water needs like irrigation systems for agriculture, as well as reservoirs, treatment plants, and distribution systems for industrial and domestic use could also be compromised. For example, after ten years of conflict in Syria, many water facilities were destroyed. As a consequence, it severely hinders the access of the population to

essential services, including water. (International Committee of the Red Cross 2021). Therefore, examining general conflict data, regardless of the reported causes, could yield insightful results. Davies, Pettersson, and Oberg 2023 did a comprehensive report on the conflict dataset done by UCDP from 1989 to 2022, reporting the organized violence in terms of “events”. They defined an event of organized violence in one calendar year as *“an incident where armed force was used by an organized actor against another organized actor, or against civilians, resulting in at least 1 direct death at a specific location and a specific date”*. Furthermore, conflict incidences were categorized into three types of organized violences: state-based armed conflicts, non-state conflicts, and one-sided conflicts, and a conflict is reported in those three cases when at least 25 people are dead in a calendar year. It is worth noting that this 25-casualty threshold implies that lower-intensity disputes and non-violent forms of conflict are not captured by the dataset, a limitation discussed further in Section 2.1.1. By analyzing the geographic and temporal distribution of conflicts, we can infer a potential relationship between water stress and conflicts, along with other variables. This data will be the main source to measure conflict and will be explained in further detail in the following chapter.

1.2.3 Foreign aid as a solution

To tackle the challenges mentioned above, a country or region might employ a range of strategies such as reforming and improving their governance, adopting innovative technologies, etc. Particularly, foreign aid, often offered by developed nations in the form of grants and loans to support the developing nations, stands out as a prominent tool. The motivation of this policy is driven by multiple factors, encompassing humanitarian concerns of improving lives for people with lesser privileges to strategic interests of exercising influence over recipient countries or fostering stability and preventing conflict spillover around the world. Major donors of foreign aid are the United States, the European Union, Japan, and China, as well as international bodies such as the World Bank, and the United Nations (OECD 2023). Foreign aid is often channelled through direct project funding. Aid projects being attributed to a country often cover a wide spectrum of sectors: government and civil services, other social infrastructure, transport and services, energy generation and supply, and banking and financial services. Among those, water-related initiatives are particularly relevant in the context of water stress. These projects may include building or upgrading critical infrastructure like water supply, sanitation, and irrigation systems, directly mitigating water stress by improving access to clean water for domestic use and boosting agricultural productivity and efficiency in water-scarce regions. Beyond infrastructure, foreign aid focuses on capacity building, enabling local communities and governments to address challenges by themselves. In the example of water-related issues, this could involve technical training in efficient water management practices, such as resource exploitation, recycling, and conservation. All of these potential contributions of foreign aid to local areas could have an effect on the onset of conflict.

1.2.4 Literature review and contribution of the paper

The literature on the impact of foreign aid shows no consensus on its effectiveness in mitigating conflict in recipient countries. Several studies indicate a positive effect, suggesting that aid can reduce conflict (Croston, Felter, and Johnston 2014; Nunn and Qian 2014; Dube and Naidu 2015; Troland et al. 2013). These authors argue that increased aid, which often are allocated to local authorities, may incentivize rival groups to sabotage aid programs, either with the goal of seizing recently arrived external resources or undermining the population’s support for the authority. Croston et al. (2014) and Troland et al. (2013)

further note that this dynamic is conditioned on the occurrence of pre-existing conflict in the area, which can expedite such sabotage efforts. On the contrary, other studies find no significant correlation between aid and conflict reduction (de Ree and Nillesen 2009; Beath, Christia, and Enikolopov 2017; Savun and Tirone 2011). These conclusions are dependent on the studied countries' circumstances, the type of aid provided, and the study's design—particularly if there are enough relevant variables to control for confounders. For instance, research by Szabó 2022 contends that the effect of foreign aid on conflict is contingent on the governance capacity of the recipient country. This perspective is complemented by Levi 2006, which summarizes the characteristics of an effective government. Together, they argue that a state can avoid conflict by leveraging its institutional strength to maintain stability, even when there are negative foreign aid shocks. Overall, these studies have one thing in common, that they often rely on cross-national or regional analyses to assess the impact of aid on conflicts. Building on this foundation, this paper seeks to investigate the effect of foreign aid on conflict in the presence of water stress at a more granular level through grid cell analysis.

In this study, India was chosen as the main country of the study. In spite of abundant monsoon rainfall and major river systems such as the Ganges and Indus, India is still ranked in the “Extreme” category at the 24th place in the national water stress rankings (Kuzma et al. 2023). India's water demand is projected to increase significantly in the coming decades, especially from the manufacturing and domestic sectors (Vaibhav et al. 2022). Moreover, even though the dependence of India's economy on foreign aid has been decreasing in recent decades, with the aid inflows-to-GDP ratio of India declining from 7 % in the 1980s to under 1 % by 2014 (Aid Data 2012), foreign aid is still a prominent tool in addressing peculiar problems in many areas in India. Along with varied types of conflict happening in India, these factors make India an ideal case for examining the influence between foreign aid and conflict under varying water stress situations. For the organization of this paper, detailed information about the methodology and harmonization process of different data as well as the descriptive statistics of the final dataset will be mentioned in chapter 2. Chapter 3 consists of the main result along with robustness tests using finer grid resolution and some external validity tests using neighboring South Asian countries. In chapter 4, a brief summary of the results and implication of the results will be included. Other secondary results and graphs will be put in the separate document *Appendix - continued*.

2 Methodology and Data

This chapter provides brief details on the data, as well as the extraction and harmonization methods in order to create final panel datasets for regressions (subsection 2.1). In the subsection 2.2, explanation about the regressions will be specified. Additionally, the project title analysis method will be mentioned as a supplementary tool for interpreting the regression results in the chapter 3.

2.1 Data

2.1.1 Conflict

As previously mentioned, the primary focus of this thesis is on the onset of conflict. To this end, we utilize detailed data from the Uppsala Conflict Data Program (UCDP) on conflict events spanning from

1989 to 2022 (Davies, Pettersson, and Oberg 2023). These events are geolocated with precise coordinates and categorized into three types: state-based, non-state, and one-sided conflicts, and they happen within a specific date within the calendar year. Each event’s intensity is also documented, including various estimates of fatalities. For the main regression model detailed in equation 1, we employ a binary dummy variable (100 or 0) to signify the occurrence of conflict. Additionally, to gain deeper insights into how foreign aid influences conflict, we explore more granular variables that differentiate between conflict types and measure conflict casualty as dependent variables. Given that the original dataset records conflicts at the event level, we aggregate this information to construct our conflict dummies and the best estimate of fatality at the grid cell level on an annual basis by using the spatial join method between the conflict spatial dataset and the grid cells of India. For this study, only the data from 1995 to 2014 will be used, because of the limited availability of the foreign aid data.

It is worth acknowledging one important limitation of the UCDP dataset: conflicts are only recorded when they result in at least 25 battle-related deaths in a calendar year. This threshold means that lower-intensity disputes, localised tensions, and non-violent forms of social conflict are excluded from the analysis. As a consequence, the results of this study pertain specifically to the onset of organized, lethal violence and may not fully capture the broader spectrum of conflict dynamics—such as demonstrations, riots, or resource-related disputes—that could also be influenced by foreign aid or water stress.

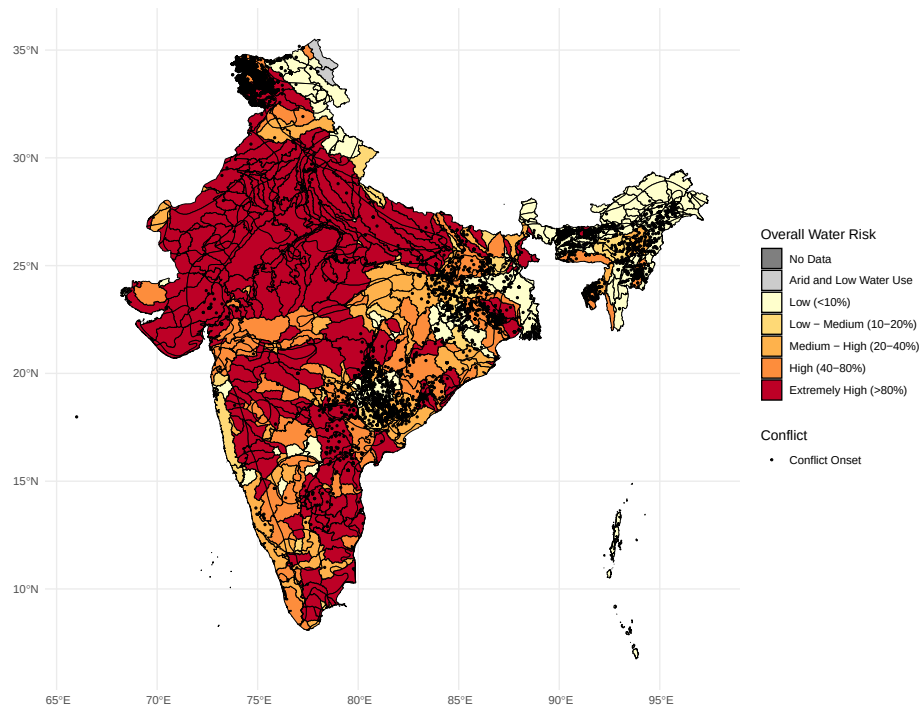


Figure 2: India’s water stress with conflict locations from 1995 to 2014 with *HydroBASINS level 6* as baseline map layer
Source: Aqueduct 4.0 by WRI and UCDP conflict dataset

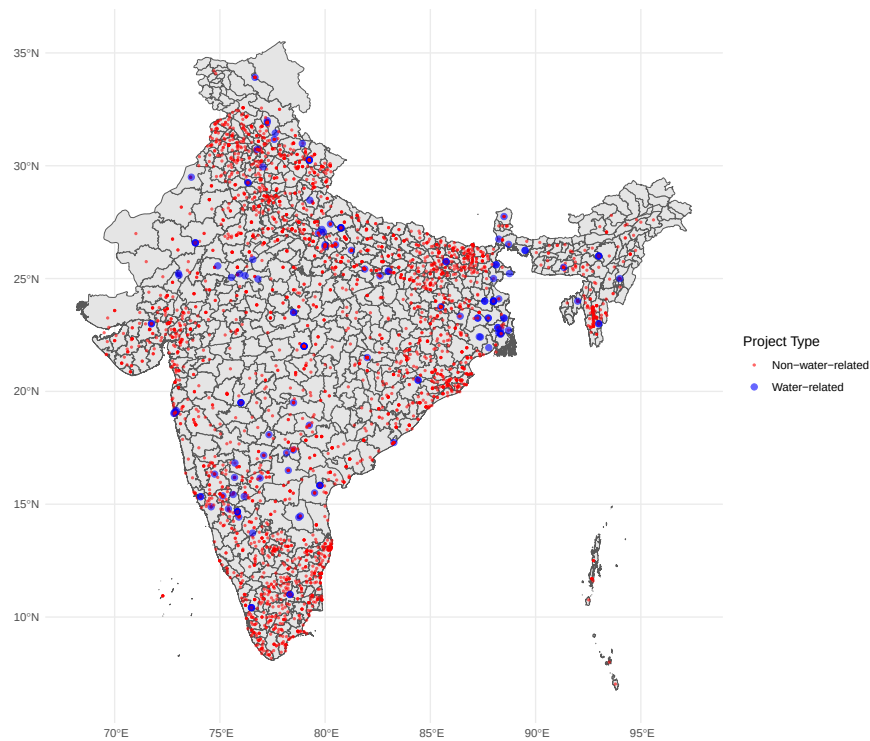


Figure 3: Distribution of World Bank Aid Project Locations from 1995 to 2014
with subdistricts (administrative level 2) as baseline map layer

Water-related sectors are agriculture, forestry, fishing, water supply and sanitation

Source: AidData — World Bank Geocoded Research Release, Version 1.4.2

2.1.2 World Bank aid projects

For this study, data on foreign aid was from the World Bank Geocoded Release by Aid Data [2017](#), which documents approved projects between 1995 and 2014 under the World Bank's IBRD/IDA lending instruments. This dataset contains precise location coordinates for each project, noting that one project might be operated in more than one location, along with the start and end years of the project's financial transactions and commitment and disbursement amounts. In our data manipulation process, we utilized the transaction start year as the starting point for each project, under the assumption that conflicts might arise as funds begin to flow into an area. For the allocation of foreign aid projects, we selected the project location count as our key variable. Although including commitment and disbursement amounts in the regressions could reveal the financial scale of aid's impact on conflict onset, our data skimming of AidData uncovered significant gaps in disbursement records. Additionally, commitment data does not accurately reflect resources that are locally delivered, because disbursements are frequently renegotiated in light of implementation realities on the ground: funds may be reallocated across sub-components, delayed due to procurement bottlenecks, or reduced following project underperformance reviews. The substantial variation in the disbursement-to-commitment ratio illustrated in figure 4 is consistent with this interpretation, showing that committed amounts regularly diverge from what is actually disbursed. Consequently, we excluded these variables from the regression analysis.

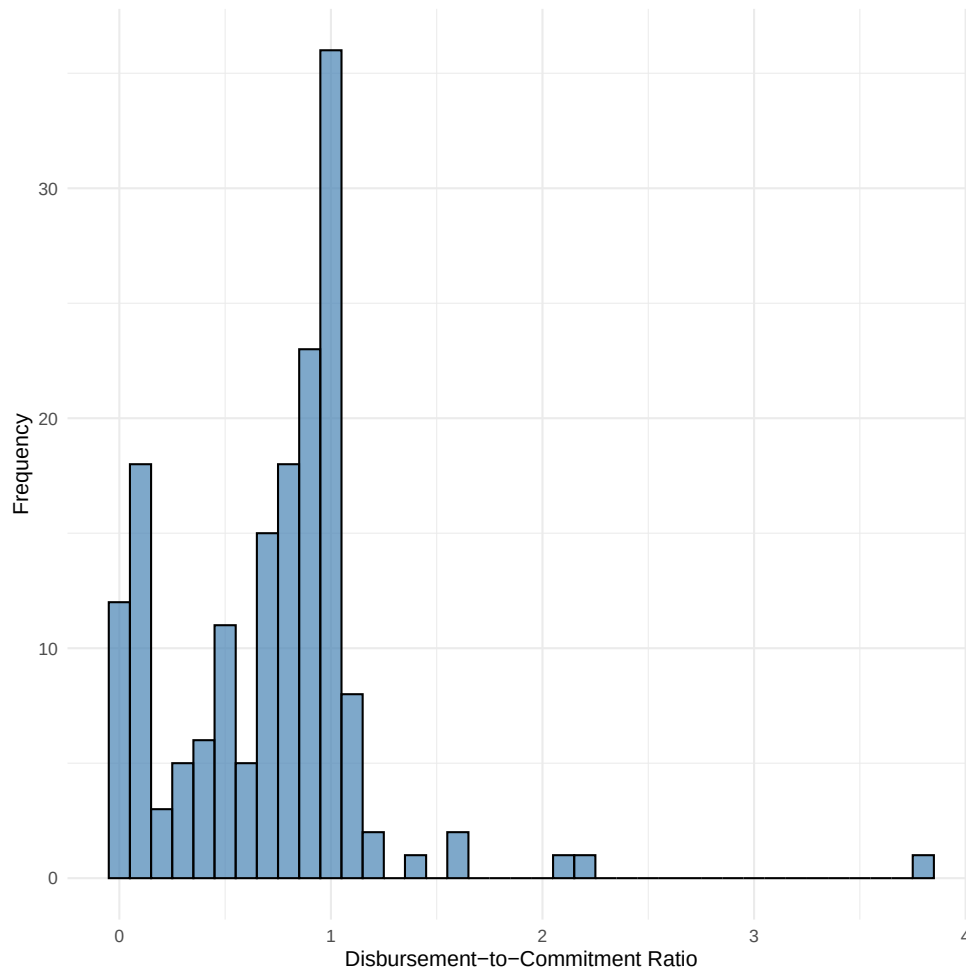


Figure 4: Disbursement-to-Commitment Ratio for World Bank Projects in India from 1995 to 2014
Source: AidData — World Bank Geocoded Research Release, Version 1.4.2

Regarding project sectors, we opted to include all projects in our analysis, irrespective of their specific sectors (see figure 5 for the distribution of project sectors across different levels of water stress). Focusing solely on aid projects related to water or any specific sector might overlook the broader socioeconomic environment that influences conflict, because water scarcity is indicated by previous research as an indirect factor to the conflict onset (UNESCO World Water Assessment Programme 2024). Likewise, including projects from all sectors allows us to consider how aid could influence local economies, governance, and social services, all of which are interlinked with the probability of conflict onset. This analysis did not include data on Chinese aid. This choice was driven by two considerations. First, Chinese aid follows different reporting standards from those of the World Bank, making a combined dataset methodologically inconsistent. Second, Chinese foreign assistance in South Asia is often structured around geopolitical objectives—such as counterbalancing Indian and US influence—rather than purely developmental goals, which raises concerns about comparability with World Bank projects (Malathi and Jayalakshmi 2025). Including both sources without accounting for these structural differences would risk conflating qualitatively distinct forms of aid. This uniformity also simplifies the methodological approach and data harmonization process.

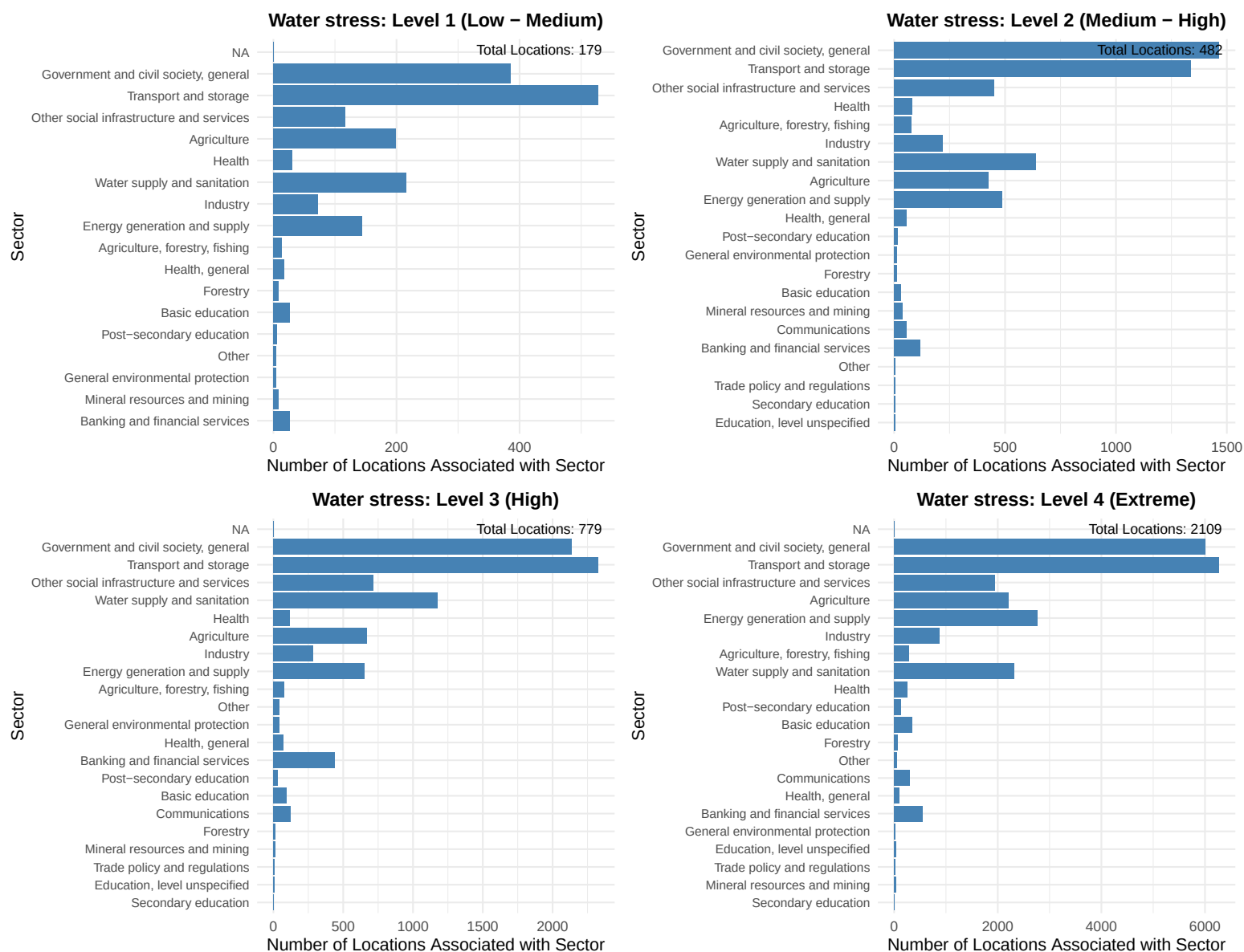


Figure 5: Distribution of World Bank Projects by Sector in India from 1995 to 2014
Note: Overlapping could happen because one project could be classified in several sectors
 Source: AidData — World Bank Geocoded Research Release, Version 1.4.2

2.1.3 Water stress

To measure the water scarcity in the region, we use the water stress index from the Aqueduct project of the World Resource Institute (WRI) (WRI 2023). This index is a part of 13 indicators in the Aqueduct project to measure the water-related risks. According to the WRI, water stress is defined as the ratio of total water demand to available renewable surface and groundwater supplies and is averaged from 1979 to 2019 in version 4 of the Aqueduct data. Its calculation was based on the hydrological sub-basin at level 6 of the HydroSHEDS classification scheme (Kuzma et al. 2023). HydroSHEDS level 6 sub-basins represent catchment areas on the order of several thousand square kilometres, which provides finer spatial resolution than national or state boundaries while still aggregating across smaller tributary systems. This scale is well-suited to the 50 by 50 km grid cell used in this study. To adapt for the baseline unit of this study, the recalibration of the index involved adjusting the value based on the weight of the surface area of HydroBASINS level 6 to that of the grid cell. Missing values in the original data were dropped. Furthermore, in some water basins, the available supply is nearly exhausted while the demand remains persistent, so the water stress index of those basins was reported as 9999 by WRI to indicate a near-complete depletion of available supply (ibid., p.11). In the dataset used in this study, a total of 11 grid cells received this top-coded value. For the consistency of the data transformation, the water stress value of 9999 is converted to the next highest normal water stress value in the dataset, which is 25.305. This recoding assigns these 11 extreme cells to the highest observed non-sentinel value, preserving their relative ranking as severely water-stressed while avoiding the distortion that a value several orders of magnitude larger than all other observations would introduce into area-weighted averaging. All 11 affected cells fall within the “Extremely High” water stress category (level 4), so this recoding does not alter their categorical classification. After the water stress value *bws_raw* transformation to grid cell is done, we constructed a categorical variable for the water stress. This variable follows the original construction of the variable *bws_cat*, which originally has 5 categories of risk—Low (0), Low-medium (1), Medium-high (2), High (3), Extremely High (4), Arid, and low water use (-1) (ibid., p.11). The category -1 was fused into category 0 because there is only one grid that falls into this category, plus it eliminates the problem of multicollinearity of potentially having other categories being removed in the regressions. The category 0 will be set as the reference for the water stress level variable. These attributed categories to each grid cell are constant across all temporal resolutions because the raw water stress value is an averaged value from the period between 1979 and 2019. As shown in the descriptive statistics in table 1, the vast majority of India’s grid cells are classified under water stress level 4 (extreme), totaling 717 grid cells, followed by level 3 (high) with 210 grid cells, level 0 (low) with 157 grid cells, level 2 (medium-to-high) with 210 grid cells, and level 1 (low-to-medium) with 48 grid cells, covering a 20-year period from 1995 to 2014.

2.1.4 Temperature

Temperature is included to capture the average thermal conditions, which are critical in understanding the long-term weather patterns that affect agricultural productivity, water availability, and consequently, the livelihoods of local populations. We pulled the temperature data from the Climatic Research Unit gridded Time Series (Harris et al. 2020). The data is in a gridded format spanning from 1901 to 2023 at a resolution of 0.5°—which corresponds to approximately 55 km per grid cell at India’s latitude, comparable to the study’s grid resolution of 50 by 50 km. We proceeded with the data manipulation by cropping the original temperature raster data to the grid cell map of India. The temperature value for each of India’s grids is then averaged from the original data of the raster. Moreover, because of the original monthly format of the temperature data, we averaged the monthly data of each year to produce

the annual temperature value for each grid. Data from the descriptive statistics table 1 indicate that the median annual temperature in India is 26.166°C, underscoring its heightened vulnerability to drought and arid conditions.

2.1.5 Precipitation

Likewise, precipitation is also an integral variable in this study. It is used to account for the water cycle's regularity and the occurrence of extreme weather events like floods and droughts. This variable provides insight into the availability of water resources, which is pivotal in regions where water stress is a known factor. High precipitation might lead to flooding, causing immediate disruption and potential conflict over damaged infrastructure or displaced populations. Conversely, low precipitation can result in drought conditions, worsening the demand for water. The used data is from the Climate Hazards Center (Funk et al. 2015). It is also in the monthly gridded satellite-based data format with very high resolution (0.05°). The data manipulation process is also similar to the one for temperature, namely that the raw data of rasters within each grid is extracted and averaged; from this, we get the single mean monthly precipitation value for each grid cell for 12 months for all recorded years. Finally, for each grid cell, those 12 monthly mean precipitation values within a given year are averaged again to produce the average monthly precipitation for a specific grid cell in a specific year, expressed in millimeters (mm). Looking at the descriptive statistic table (table 1), India has the median average monthly precipitation of 86.44 mm, indicating moderate conditions for most grid years. The range from 3.264 mm (minimum) to 446.451 mm (maximum) highlights the presence of both extremely dry and extremely wet grid years, reflecting India's diverse climate. And with the moderate variation of 61.4121 mm for the interquartile value despite the extreme difference between maximum and minimum values, there is a good argument for including precipitation in our study's regressions as a control variable.

2.1.6 Cropland percentage

To control for the agricultural production and its land use within each grid cell, we incorporate cropland percentage as a control variable. According to the data source from Potapov et al. 2022, cropland percentage is defined as the percentage of land used for annual and perennial herbaceous crops for human consumption, forage (including hays). By including the proportion of land dedicated to crop production, we isolate and control for the socio-economic implications of agriculture on conflict dynamics, because increased agricultural activity could secure the food demand—reducing conflict potential or overstressing local resources, leading to conflict. This control is crucial since agriculture accounts for more than 70% of freshwater withdrawal (Vaibhav et al. 2022, p.43), and its demand is projected to even further increase in the future, directly linking it with temperature and precipitation variables that influence flood and drought risks.

The raw raster data has a very high resolution of approximately 1 km. It is available in four-year interval epochs, from 2003 to 2019, which makes it 5 observation points. Here we take the cropland percentage value for the beginning year of each epoch to be equal to the average of the epoch, i.e., the cropland percentage for 2003 is the average value of the epoch 2000-2003. This approach assumes that cropland percentage change is relatively stable within each epoch, which simplifies the extrapolation process. It is acknowledged that this method could smooth out the relative fluctuations that occurred during these periods, but because of the limitation of the raw data, we proceeded as mentioned above.

Similarly to the two previous variables, we extracted and then applied the mean function to find

out the cropland percentage value of each grid cell. Due to the limit of temporal observations (only 5 observation points), we tried to extrapolate the data to get the individual annual value of the cropland percentage from 1995 to 2014. Firstly, via the raw data, we calculated the four growth rates between each original observation point. The annual growth rate is then converted by taking the fourth root of the compound growth factor, since there are 4 years in each interval. To estimate cropland percentages for the years before the first data point of 2003, we used the converted annual growth rate in reverse. When looking at the literature review on the agricultural expansion of India, we see that the rate of change in cropland percentage was generally lower in the decades before the turn of the century due to less intensive mechanization, slower adoption of high-yield varieties, and different government policies focusing on land reforms rather than expansion. (Singh and Bhalla 2009). Therefore, we assume that the annual growth rate of cropland was about half or less of what we observed in the 2000s for the backward extrapolation. For the forward extrapolation from 2003 onwards, we used linear interpolation to estimate the cropland percentage based on the five known data points. In the descriptive statistic table (table 1), the cropland percentage of India's grid cells ranges from 0 % to 98.1 %, with a median of 35.8 % and an interquartile range of 50.9 %. This demonstrates a moderate but highly heterogeneous agricultural presence across India's territory. Indeed, India's diverse land use, from non-arable to intensive farming zone, makes a strong case for including the cropland percentage variable as one of the control variables. This reinforces the linkage between agricultural activity, water use, and conflict.

2.1.7 GDP

To control for economic activities and development within each grid, we utilized gridded GDP data from Gaffin et al. 2004, available via the Socioeconomic Data and Applications Center (SEDAC). This dataset provided GDP predictions for 2025 with a 1990 baseline, based on the Special Report on Emissions Scenarios (SRES) B2 scenario, which focuses on local solutions for sustainability, environmental protection, and social equity with slower technological and economic growth (Nakicenovic and Swart 2000). We extracted and recalibrated the GDP data for 1990 and 2025 to match our study's grid cells. Importantly, the result of this extraction process is the mean GDP for each grid cell, providing an average economic output per unit area. Additionally, we used a complementary dataset from the same source (SEDAC) to derive national annual growth rates from 5-year interval GDP predictions (1990-2100) under the B2 scenario (Findley et al. 2011). These deducted annual growth rates are then used to apply to the ensemble of the baseline grids' GDP for the extrapolation step. Due to the limited temporal coverage of the gridded data, we assumed homogeneous growth across all grids based on these national rates. For extrapolation, we created a comprehensive data frame for each grid cell, adjusting zero GDP values in 1990 to a minimal non-zero value to ensure computational consistency. We then apply the national annual growth rates generated above to the grid cell's initial GDP value of 1990 to project GDP values year by year up to 2025. Grid cells with zero GDP in both 1990 and 2025 are maintained at zero for all those in-between years. Finally, the extrapolated dataset is filtered to include only data from 1995 to 2014.

2.1.8 Population

Similar to the GDP data, we utilize gridded population counts from SEDAC (Center For International Earth Science Information Network-CIESIN-Columbia University 2018), at the resolution of 30 arc-seconds (roughly 1 km by 1 km at the equator). This data aggregates national census information from 2000 to 2020 in 5-year intervals through a gridding algorithm. Including population as a control variable is crucial for understanding the increased demand for water for domestic, industrial, and auxiliary uses,

such as cooling in power plants, alongside population expansion. The extraction method of the raw data onto the baseline unit of this study is identical to the one of GDP data, by projecting the data from the original rasters to each grid cell and then calculating the mean of the original raster data within each grid cell. This approach yields the population density for each grid. As the aggregated data only has 5 observations for each grid from 2000 to 2020, we proceeded to extrapolate the population density data from 1989 to 1999 using a decreasing growth rate based on historical trends from UN data for India, accounting for the observed decreasing growth trend in this period (Department of Economic and Social Affairs Population Division 2024). From 2000 onwards, we employ linear interpolation between known census years (2000, 2005, 2010, 2015, and 2020) to fill in the population data gaps. Then, the final extrapolated data is filtered to include only the temporal resolution from 1995 to 2014.

2.1.9 Descriptive Statistics

	Descriptive Statistics							
	Obs	Mean	Std	Min	Q1	Median	Q3	Max
Conflict dummy	25340	8.074	27.244	0.000	0.000	0.000	0.000	100.000
State-based conflict dummy	25340	6.066	23.870	0.000	0.000	0.000	0.000	100.000
Non-state conflict dummy	25340	0.608	7.772	0.000	0.000	0.000	0.000	100.000
One-sided conflict dummy	25340	4.365	20.431	0.000	0.000	0.000	0.000	100.000
Project location count	25340	0.150	0.593	0.000	0.000	0.000	0.000	13.000
Best death estimate in log	25340	0.157	0.626	0.000	0.000	0.000	0.000	6.248
Water stress level 0	3140	0.034	0.083	0.000	0.002	0.011	0.058	0.098
Water stress level 1	960	0.150	0.031	0.100	0.126	0.152	0.177	0.199
Water stress level 2	2700	0.302	0.060	0.201	0.247	0.303	0.354	0.399
Water stress level 3	4200	0.583	0.119	0.402	0.482	0.581	0.677	0.799
Water stress level 4	14340	4.816	5.486	0.801	1.310	2.263	6.125	25.305
GDP in log	25340	1.641	0.764	0.000	1.244	1.642	2.115	5.292
Population in log	25340	5.055	1.285	-0.866	4.729	5.218	5.779	9.161
Average temperature	25340	24.605	5.829	-7.040	25.146	26.166	27.096	30.093
Average monthly precipitation	25340	95.586	57.487	3.264	57.839	86.444	119.260	446.451
Cropland percentage	25340	0.389	0.288	0.000	0.125	0.358	0.634	0.981

Table 1: Descriptive statistics of Key Variables
Annual data for India's grid cells from 1995 to 2014

2.2 Methodology

$$P\{\text{Conflict}_{it} = 100\} \text{ or } \text{Conflict}_{it} = \beta_1 \text{Aid}_{it} + \beta_2 (\text{Aid}_{it} \times \text{Water Stress}_i) + \beta_3' \mathbf{X}_{it} + \eta_{jt} + \tau_i + \epsilon_{it} \quad (1)$$

The equations 1 above is the main regression equations of the study; the first dependent variable is for all conflict dummy variables, and the second dependent variable is for the conflict casualty variable. The identification of the study is being done at the grid cell level i by arbitrarily choosing the dimension of 50 by 50 km. After being created, its centroids are projected on the country map. Only the grids with the centroids being located inside the country's territory are then kept as the unit of the study (see figure 6). It is understandable that there would be grid cells that are not totally inside one country's territory;

for example, they could entrench into the neighboring countries or the ocean. These differences among grid cells could be controlled via the inclusion of the grid cell fixed-effect. Furthermore, as a robustness check in chapter 3.2, we will utilize smaller grid cell dimensions as the base unit of our study. This adjustment aims to reduce the instances where grid cells extend into neighboring countries or over the ocean, thereby enhancing the precision of our geographical analysis. External validity tests were also conducted on some neighboring South Asian countries; the aim was to observe whether India's result pattern is applicable to its neighboring countries. And all the data for each variable are harmonized at the annual level t , as discussed in the subsection 2.1.

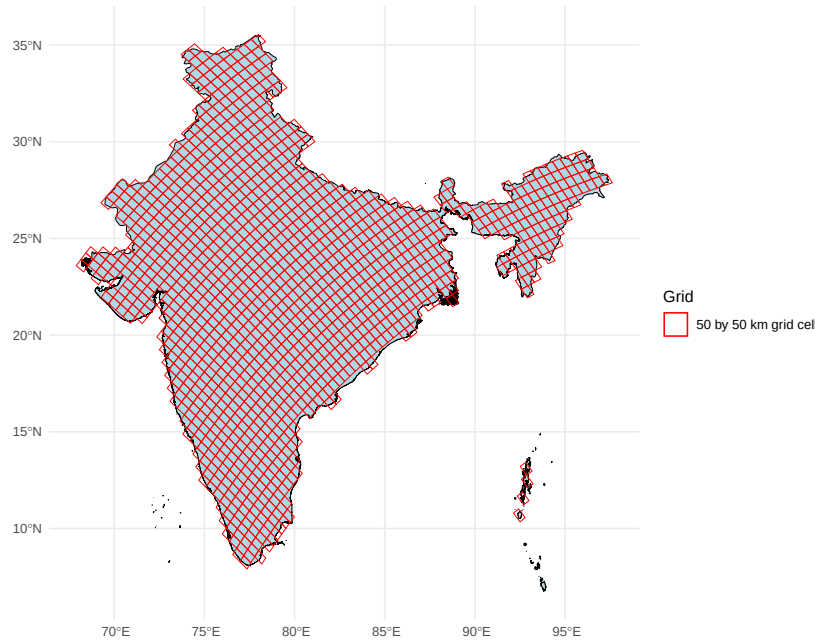


Figure 6: 50 by 50 km grid cells in India

The assumption in the equation is that foreign aid could have an effect on the onset of conflicts. Moreover, by allocating foreign aid projects in the water-scarce areas, the existing impact of foreign aid on conflict could be even further amplified or shift in the other direction; hence, the interaction term of foreign aid and water stress variables is included. Besides, control variables $\mathbf{X}$ such as the logarithm of GDP mean, the logarithm of population density, average temperature, average monthly precipitation, and cropland percentage were included to account for potential confounding factors like economic development, demographic pressures, climatic conditions, and agricultural land use that could otherwise obscure or bias the estimated relationship between foreign aid projects and conflict, especially in water-stressed areas. Because the final dataset used is a panel data, we have to control for the time- and entity-fixed effects. Here, we choose to take into account the grid cell fixed effect (denoted τ_i), capturing the time-invariant difference between grid cells, and the regional time trend fixed-effect capturing the dynamic of subdistrict characteristics evolving over time (denoted η_{jt}). Finally, the standard error is clustered at the grid cell i level in order to take into consideration heteroskedasticity and for arbitrary autocorrelation within an entity but treat the errors as uncorrelated across grid cells (Watson and Stock 2018, p.376).

Noticeably, the regression model specified in equation 1 employs a linear probability model (LPM)

when the conflict-related dependent variable is binary, taking values of 0 or 100. This approach facilitates straightforward interpretation of the coefficients. However, treating the binary dependent variable as continuous in the LPM can result in predicted probabilities falling outside the $[0,100]$ range. This contrasts with probit or logit models, which constrain prediction values within the predefined range. As a solution, predicted values below 0 can be interpreted as 0, and those above 100 as 100. Conversely, when the conflict-related dependent variable is continuous in equation 1, the regression results provide insights into how foreign aid affects the intensity of conflicts, namely the best casualty count. Regarding the unit of study, using grid cells as the primary unit differentiates this research from other studies on the same topic. While many studies of the same topic might adopt a regional or cross-national approach, analyzing at the grid cell level could reveal how geographical and environmental factors, like water stress, interact with aid at a very specific level, enhancing our understanding of aid's impact on conflict dynamics.

Because the project-related variable of our study is only the project location count in each grid cell, to better interpret this variable and its interactions with different water stress levels, we will also do a separate project title analysis as well as look at the project's sector distribution for each level of water stress. To do the project's title analysis, individual project titles for each water stress category will be scanned, categorized into key themes, and examined to identify patterns. It is acknowledged that one could just use the already existing *ad_sector_codes* variable of AidData dataset that is attributed to each project. Nevertheless, we argue that looking at both the project titles and its default sector distribution will provide more nuances and insights into the nature of an aid allocated into an area, like the fact that a project could be an emergency measure for the distressed situation of an area, or a project could be aimed within a small-scale and narrowly-defined goal. This element is not documented in the sector code but rather in the project title itself. The categorization of the project title will be done as follows:

- Agriculture: Projects supporting agricultural development and food security.
- Community cohesion: Projects that are focused on strengthening the social linkage of the community.
- Education: Projects aiming at improving educational access and quality.
- Emergency Response: Projects addressing disaster recovery and emergency relief.
- Finance: Projects related to financial services, microfinance, and economic development.
- Governance: Projects improving governance, legal systems, and public administration.
- Health: Projects focusing on healthcare, nutrition, and disease prevention.
- Infrastructure: Projects involving transportation, energy, and urban development.
- Security: Projects that are involved with peacebuilding and conflict resolution.
- Small scale: Projects that are focused on small, rural communities.
- Water Management: Projects relating to water supply, sanitation, irrigation, and resource management.

For each category, several keywords are associated for the scanning. If one project's title matches one keyword in a particular category, then the project is assigned to that category. In case a project's title is matched to several categories, the count will be done more than once accordingly. This matches the categorization method of AidData where one project could be in several sectors. The graphs from the project title analysis will be provided in the section 6. Sector distribution graphs for each water stress category are provided in figure 5.

3 Analytical results

3.1 Main results

3.1.1 Conflict Dummy

50 by 50 km grid cells	
Dependent Variable:	Conflict Dummy
Model:	(1)
<i>Variables</i>	
Project Location Count	2.983* (1.625)
Project Location Count $\times$ Water Stress Level 1	-5.882* (3.018)
Project Location Count $\times$ Water Stress Level 2	-2.131 (2.357)
Project Location Count $\times$ Water Stress Level 3	-2.734 (1.721)
Project Location Count $\times$ Water Stress Level 4	-2.944* (1.692)
Log GDP	-38.81*** (13.45)
Log Population	5.899 (21.22)
Average Temperature	8.154 (5.191)
Average Precipitation	0.0538 (0.0380)
Cropland Percentage	-6.992 (12.33)
<i>Fixed-effects</i>	
Grid Cell	Yes
Subdistrict * Year	Yes
<i>Fit statistics</i>	
Observations	25,340
R ²	0.79814
Within R ²	0.00381
<i>Clustered (grid_id) standard-errors in parentheses</i>	
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>	

Table 2: Main result with conflict dummy as the dependent variable for India

The primary finding of the study is presented in Table 2, where the dependent variable, a conflict dummy, takes a value of 0 or 100 to indicate the absence or presence of conflict, respectively. The nature of

the foreign aid is that one project could be operated within more than one location. Therefore, the variable “project location count” accounts for the number of total locations of projects within each grid, no matter if they are under the umbrella of one project or not. At 10 % of statistical significance, one additional aid project location in a grid cell with no water stress (the base category, level 0) increases the probability of conflict happening by 2.983 percentage points. However, because the regression includes interaction terms between project location count and water stress levels, the coefficient on the standalone project location count should be interpreted as the effect specific to level 0 areas only. For grid cells with different water stress levels, the total marginal effect of an additional project location must account for both the main coefficient and the relevant interaction term. Projects located in areas with low-to-medium water stress (level 1) reduce the net probability of conflict, with a total marginal effect of $2.983 - 5.882 = -2.899$ percentage points (both coefficients significant at 10%). Similarly, in extreme water stress areas (level 4), the net effect is $2.983 - 2.944 = 0.039$ percentage points, with the interaction coefficient significant at 10%. These results indicate that well-targeted aid can mitigate conflict in water-constrained settings. In low-to-medium water stress areas of India, projects could prevent conflict by enhancing resilience through small-scale interventions like irrigation, fostering community cohesion, and avoiding competition over resources. Looking at the actual projects being allocated in water stress level 1 areas, we confirm the prevalence of the above-mentioned purpose among these 66 projects (see figure 7). With both water-related and non-water-related projects, one can observe the efforts on solving the basic water problems as well as a broader attempt to stabilize the areas with projects in infrastructure, health, and education. This aligns with CEPR’s findings that aid is effective in stable contexts with a narrowly defined goal that helps to promote social stability (Troland et al. 2013).

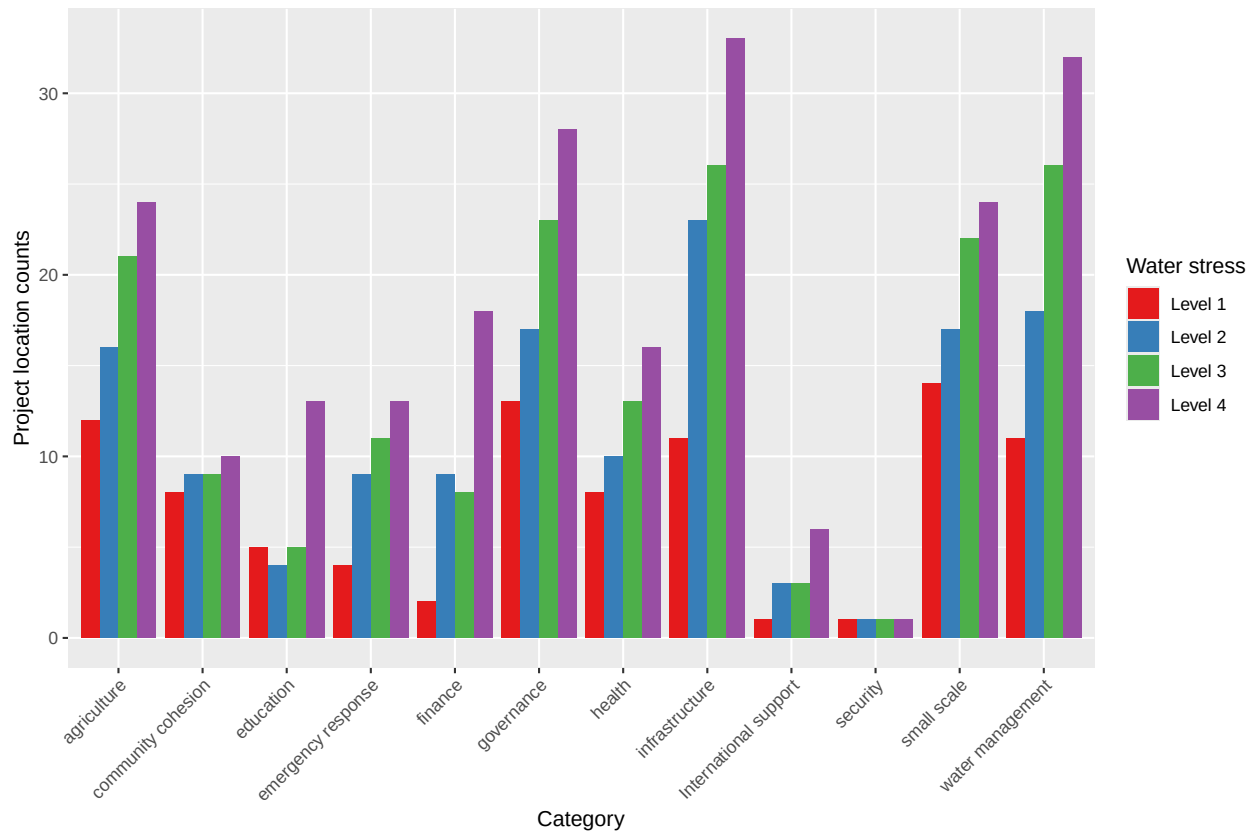


Figure 7: Project Title Analysis for India

Note: one project could be categorized in more than one category and could have more than one location

In extreme water-stressed areas (level 4), the net marginal effect of an additional aid project location on conflict probability is nearly zero ($2.983 - 2.944 = 0.039$ percentage points), suggesting that aid neither meaningfully reduces nor amplifies conflict in the most severely constrained settings. This near-neutral result is notable: it implies that aid projects manage to offset almost entirely the baseline conflict-inducing tendency observed in level 0 areas, but fall short of achieving the clear pacifying effect seen in level 1. Among the 193 projects allocated to these areas, a high concentration of water management and governance projects can be observed (figure 7), suggesting that project design is responsive to the severity of scarcity on the ground. However, the complexity of extreme water stress environments—marked by deep institutional failures, entrenched resource competition, and populations whose survival needs are already severely compromised—may limit what aid projects alone can achieve. As Michel (2024) argue, water stress acts as an indirect conflict driver whose effect is mediated by governance capacity, power asymmetries, and structural inequalities; in areas where all of these conditions are simultaneously adverse, aid projects face a particularly constrained operating environment. This is consistent with the broader finding in the literature that, absent strong institutional conditions, aid tends to have no statistically significant effect on conflict onset (de Ree and Nillesen 2009). Conversely, the non-significant effects in medium-to-high (level 2) and high (level 3) water stress areas suggest that aid’s impact is also limited there, potentially because those zones combine elevated resource competition with insufficient institutional capacity to channel aid effectively. Water-related projects in those areas target diverse populations spanning urban water supply, rural sanitation, and agriculture, thereby risking the intensification rather than resolution of distributional disputes. Furthermore, large-scale projects such as dams and irrigation schemes can create winners and losers in water distribution, further complicating

local dynamics (Michel 2024). Projects in states like Andhra Pradesh, Tamil Nadu, and Karnataka that are known for water governance challenges are likely to face these issues, making allocated foreign aid projects less effective due to the complexity on the ground.

Among the control variables, the logarithm of GDP is significant at 1 % level, showing that a 1 % increase in GDP will decrease the chance of conflict onset by 0.3881 percentage point, underscoring the crucial role of economic development in stabilizing India and directly reducing conflict risk during this period. One might ask whether the foreign aid itself contributes to the economic growth of India. In reality, India has been less dependent on foreign aid over time when verifying the aid-GDP ratio. Nevertheless, aid can still play a targeted role in specific sectors or regions, as the result told us above. In the regression function, the GDP and foreign variables are both included. Therefore, one could attribute the statistical significance effect of GDP on mitigating conflicts to India's own efforts on industrialization, technological advancements, and a burgeoning service sector.

3.1.2 Other auxiliary dependent variables

50 by 50 km grid cells Dependent Variables: Model:	State-based dummy (1)	One-sided dummy (2)	Non-state dummy (3)	Log Best Death Count (4)
<i>Variables</i>				
Project Location Count	2.816** (1.380)	2.667 (2.165)	-0.6337 (0.4948)	0.0973 (0.0718)
Project Location Count $\times$ Water Stress Level 1	-4.717 (2.906)	-5.696* (3.252)	1.206 (0.9039)	-0.1680* (0.0940)
Project Location Count $\times$ Water Stress Level 2	-2.511 (1.993)	-1.038 (2.581)	0.5357 (0.5054)	-0.0803 (0.0737)
Project Location Count $\times$ Water Stress Level 3	-2.256 (1.541)	-2.775 (2.140)	0.4259 (0.4995)	-0.0791 (0.0702)
Project Location Count $\times$ Water Stress Level 4	-2.264 (1.457)	-3.182 (2.171)	0.4896 (0.5175)	-0.0921 (0.0714)
Log GDP	-36.40*** (11.78)	-21.24* (10.99)	-2.335 (1.541)	-0.8222*** (0.2761)
Log Population	-3.120 (14.75)	14.39 (16.14)	11.85 (8.264)	0.1135 (0.4300)
Average Temperature	3.391 (5.143)	4.508 (4.819)	0.3826 (1.239)	0.2018 (0.1301)
Average Precipitation	0.0345 (0.0347)	0.0243 (0.0271)	-0.0071 (0.0086)	0.0008 (0.0007)
Cropland Percentage	-8.564 (10.88)	-8.743 (9.001)	1.064 (2.698)	-0.2358 (0.2143)
<i>Fixed-effects</i>				
Grid Cell	Yes	Yes	Yes	Yes
Subdistrict * Year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	25,340	25,340	25,340	25,340
R ²	0.78199	0.76908	0.82035	0.85987
Within R ²	0.00367	0.00312	0.00210	0.00550

Clustered (grid_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Regressions with other auxiliary variables of conflict as dependent variables for India

The table 3 categorizes conflict into three subcategories—state-based, non-state-based, and one-sided violence—as defined by Davies, Pettersson, and Oberg 2023. We constructed three dummy variables to indicate the occurrence of each violence type: each variable takes a value of 0 if no incident occurs and 100 if an incident occurs, enabling a clear assessment of conflict prevalence. Besides that, the logarithm of the best death estimate was also tested as a dependent variable to measure the magnitude of the incidence. Looking at the results, we see that foreign aid does affect the onset of state-based conflict in areas with no water stress (the base category), with the coefficient of 2.816 at 5% of statistical significance. Once an aid project is introduced into local areas, the resources available to the local government increase. This shifts the power dynamics between the state and excluded groups. Coupled with the diverse governance capability in a country like India, rebel groups could escalate conflicts to challenge the state out of grievance of being excluded. On the other hand, the interaction terms between project location count and any water stress level are not statistically significant for state-based conflict, meaning that the water stress context does not meaningfully modify this dynamic. Similarly to the main result using the conflict dummy, the logarithm of GDP heavily influences the outcome of state-based conflict, with a 1% increase in GDP lowering the probability of state-based conflict onset by 0.3640 percentage point (significant at 1%).

As for non-state conflict, neither the standalone project location count nor its interaction terms with any water stress category reach statistical significance. This indicates that conflicts between non-state actors are not influenced by aid allocation, likely because they are driven by internal group dynamics rooted in existing ethnic and religious differences or prior territorial and power disputes. For the dummy of one-sided conflict, the standalone project location count coefficient is 2.667 but is not statistically significant. Only projects allocated in low-to-medium water stress areas (level 1) show a significant effect, with an interaction coefficient of -5.696 (significant at 10%). The net marginal effect of an additional project location in level 1 areas is therefore $2.667 - 5.696 = -3.029$ percentage points, indicating a conflict-reducing outcome. Similar to the main result, local areas with water stress level 1 do not face alarming resource scarcity, so the presence of aid projects could ease the desperation of resource constraint among groups that might otherwise target civilians for resource grabs. Beyond this level of resource constraint (level 2 and above), the aid projects might be insufficient to address the deeper scarcity. Moreover, GDP also contributes to mitigating the onset of one-sided conflict, though at a lower magnitude when compared to state-based conflict, with a 1% increase in GDP lowering the probability by 0.2124 percentage point (significant at 10%).

Finally, looking at the logarithm of the best death estimate as the dependent variable in model (4), we observe that the standalone project location count has a weakly significant positive coefficient of 0.0973 (at 10%), consistent with the main result that aid in level 0 areas slightly raises conflict-related outcomes. However, projects allocated in low-to-medium water stress areas (level 1) generate a net reduction in conflict fatality: the interaction coefficient of -0.1680 (significant at 10%) outweighs the standalone effect, yielding a net marginal effect of $0.0973 - 0.1680 = -0.0707$, corresponding to an approximate 7.07% reduction in the fatality count. Economic development also reduces conflict fatality, with a 1% increase in GDP associated with a decrease of 0.8222% in the best death estimate (significant at 1%).

The main results demonstrate a complex picture of the foreign aid–conflict dynamic under water

stress in India by breaking down the initial conflict dummy into its three subcategories of violence dummy variables. The conflict onset depends very much on the local actors at play, the resource situation and power dynamic, as well as the water stress level of local areas in some situations. In the next subchapter, we will present the main regressions again, but by modifying the baseline unit of the study to a 25 by 25 km grid cell to serve as robustness tests.

3.2 Robustness tests

25 by 25 km grid cells for India					
Dependent Variables: Model:	Conflict Dummy (1)	State-based dummy (2)	Non-state dummy (3)	One-sided dummy (4)	Log Best Death Count (5)
<i>Variables</i>					
Project Location Count	0.9662 (0.9129)	1.010 (0.8185)	0.0346 (0.2986)	0.7552 (0.9356)	0.0607* (0.0332)
Project Location Count × Water Stress Level 1	-0.5727 (1.602)	-2.447 (1.516)	0.8778 (0.9011)	-1.490 (1.490)	-0.0821** (0.0413)
Project Location Count × Water Stress Level 2	-1.784 (1.222)	-2.449** (1.090)	0.1074 (0.3165)	-0.3276 (1.067)	-0.0725** (0.0348)
Project Location Count × Water Stress Level 3	-1.025 (1.096)	-0.7143 (0.8926)	-0.2510 (0.3496)	-0.5931 (1.067)	-0.0639* (0.0342)
Project Location Count × Water Stress Level 4	-1.141 (0.9719)	-1.033 (0.8419)	-0.2677 (0.3455)	-0.7996 (0.9491)	-0.0598* (0.0336)
Log GDP	-13.05*** (3.326)	-7.184** (2.889)	-2.456*** (0.9392)	-7.799*** (2.513)	-0.2145*** (0.0749)
Log Population	-1.771 (5.839)	-1.780 (4.857)	-0.3136 (2.500)	-0.1683 (3.343)	-0.0640 (0.1069)
Average Temperature	1.983 (1.680)	0.2922 (1.494)	-0.0964 (0.4083)	0.9487 (1.211)	0.0527* (0.0301)
Average Precipitation	0.0358*** (0.0100)	0.0315*** (0.0087)	-0.0006 (0.0022)	0.0123 (0.0079)	0.0006*** (0.0002)
Cropland Percentage	-4.823** (2.436)	-1.336 (1.899)	-0.1928 (0.4456)	-3.430* (1.809)	-0.0428 (0.0403)
<i>Fixed-effects</i>					
Grid Cell	Yes	Yes	Yes	Yes	Yes
GID_2-year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	100,880	100,880	100,880	100,880	100,880
R ²	0.56642	0.55901	0.48972	0.48750	0.67022
Within R ²	0.00080	0.00057	0.00052	0.00045	0.00099

Clustered (grid_id) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 4: Robustness check for regression results of India

The table 4 above presents the robustness check of the main results by choosing 25 by 25 km grid cells as the unit of study. This grid is four times smaller than the original grid cell of 50 by 50 km. Consequently, this adjustment yields more observations—now exceeding 100,000 compared to roughly 25,000 with the 50 by 50 km grids. The structure of some key variables also shifts, as we now focus on units that better capture micro-level variations in water stress, aid project locations, and conflict incidents.

The results differ markedly from the 50 by 50 km findings when the conflict dummy or its sub-categories are used as dependent variables. For the general conflict dummy (model 1), the state-based dummy (model 2), the non-state dummy (model 3), and the one-sided dummy (model 4), the standalone project location count is not statistically significant. The only exception is in model (2), where projects allocated in medium-to-high water stress areas (level 2) show a negative and significant interaction coefficient of -2.449 (at 5%), indicating a net conflict-reducing effect of $1.010 - 2.449 = -1.439$ percentage points for state-based conflict in those specific areas. As in the 50 by 50 km analysis, the logarithm of GDP retains a negative and significant effect on conflict onset across all models. Interestingly, additional control variables also become statistically significant at this finer resolution. Cropland percentage in models (1) and (4) induces a negative effect on conflict onset, indicating that a well-spread agricultural landmass and production could reduce tensions by providing food security to civilians in local areas. Average monthly precipitation in models (1) and (2) has a positive and significant effect on conflict onset, consistent with the interpretation that excessive rainfall increases flood risk, leading to population displacement, resource competition, and social tensions that can spark conflict—a pattern appropriate for India’s monsoon-prone regions.

Model (5) with the logarithm of the best death estimate as the dependent variable offers the most complete picture of the aid–conflict dynamics under water stress at the finer spatial resolution. The standalone project location count is positive and significant at 10%, with a coefficient of 0.0607, consistent with the 50 by 50 km result. The interaction terms with water stress levels 1 through 4 are all negative and statistically significant, with coefficients of -0.0821 , -0.0725 , -0.0639 , and -0.0598 respectively. In each case the net marginal effect is negative except the extreme water stress level 4. The GDP variable again exerts a strong negative influence, with a 1% increase in GDP associated with a 0.2145% reduction in the fatality count (significant at 1%). Temperature and precipitation are also positively and significantly associated with conflict fatality, consistent with the interpretation that heat and flood risks heighten resource competition and aggression. While the main result from the 50 by 50 km analysis—showing a significant effect of aid projects on the general conflict dummy—was not replicated at the finer resolution, the robustness test offers a compelling complementary insight: at finer spatial scales, aid may not consistently determine whether conflicts arise, but it plays a critical role in reducing their severity when they do. The finer spatial scale also uncovers nuanced environmental influences on conflict dynamics through the control variables of temperature, precipitation, and cropland percentage.

3.3 External validity

50 by 50 km grid cells				
Dependent variable: Conflict Dummy				
Country:	Afghanistan	Bangladesh	Nepal	Pakistan
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Project Location Count	20.53 (23.39)	-3.065*** (0.9285)	-3.883** (1.530)	3.379 (2.539)
Project Location Count × Water Stress Level 1	-21.24 (22.62)	12.39*** (2.545)	2.422 (1.778)	-6.223* (3.190)
Project Location Count × Water Stress Level 2	-28.38 (24.89)	3.404 (3.672)	4.230*** (1.472)	-4.590 (3.399)
Project Location Count × Water Stress Level 3	-16.68 (26.58)	4.877** (2.274)	1.477 (2.058)	-4.196 (2.929)
Project Location Count × Water Stress Level 4	-23.45 (23.75)	4.267* (2.213)	0.7950 (2.516)	-1.304 (2.851)
Log GDP	-23.45 (61.33)	-174.2 (105.8)	-29.82 (30.01)	47.29*** (15.76)
Log Population	20.80 (40.80)	47.75 (111.6)	4.210 (9.961)	15.24 (10.37)
Average Temperature	-0.6177 (23.74)	-87.96 (91.03)	2.496 (27.31)	0.0933 (3.343)
Average Precipitation	2.219** (0.9119)	-0.1297 (0.3519)	0.2614** (0.1177)	0.3479*** (0.1037)
Cropland Percentage	487.7 (503.5)	229.7 (259.6)	85.65 (176.5)	-25.60 (27.28)
<i>Fixed-effects</i>				
Grid Cell	Yes	Yes	Yes	Yes
Subdistrict * Year	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,160	1,100	1,260	7,000
R ²	0.88559	0.90069	0.76128	0.55020
Within R ²	0.01950	0.09459	0.01802	0.01454

Clustered (grid_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: External validity with other South Asian countries

To assess the external validity of the study, we run our models on the neighboring countries of India in the South Asia region. Most countries in South Asia also have the problem of water stress. And besides unique cases like Bhutan or the small island of the Maldives, all the other countries are flooded with violence and conflicts. At the same time, those countries also receive flows of foreign aid across many different sectors. The region of South Asia consists of eight countries, including India, but only four

countries were chosen to be reported in the table 5—Afghanistan, Bangladesh, Nepal, and Pakistan. Sri Lanka was excluded in the 50 by 50 km grid analysis because of multicollinearity; Bhutan and the island of the Maldives do not have many conflict incidents to be qualified for the external validity tests. Robustness checks that are run at a final spatial resolution of 25 by 25 km will be reported in the separate document **Appendix - continued**. For the interpretability of the variable project location count, we look at the project sector distribution for each water stress level based on the variable *ad_sector_codes* and the project title analyses. These graphs are available in the separate document **Appendix - continued**.

Overall, the results from these four countries reveal different patterns when compared to those observed in India. Looking at Afghanistan, aid project locations and allocation exhibit no statistically significant effects on conflict onset in the 50 by 50 km grid analysis—neither the standalone project location count nor any of its interaction terms with water stress levels reach conventional significance thresholds. This finding is unsurprising in a war-torn country like Afghanistan, where chronic conflicts driven by political instability, ethnic tensions, and geopolitical influences likely overpower any detectable contribution of foreign aid projects or water stress dynamics. The country’s prolonged instability creates substantial logistical and institutional barriers that could diminish aid effectiveness at this coarser grid cell resolution. Furthermore, the 50 by 50 km grid cell may aggregate data across broad areas, thereby obscuring local variations in how aid influences conflict. However, when a robustness check for the external validity test (see subsection 2.2.1 in the separate document **Appendix - continued**) is conducted using a finer 25 by 25 km grid, aid projects begin to show a discernible effect on conflict dynamics, suggesting that the influence of foreign aid on conflict in Afghanistan is more distinguishable at a more granular level.

The result for Bangladesh depicts a pattern that partly mirrors and partly inverts India’s findings. The standalone project location count significantly reduces conflict onset by 3.065 percentage points (significant at 1%), consistent with India’s finding that aid in low-stress base areas can be stabilising. However, the interaction terms tell a starkly different story. Projects allocated in low-to-medium water stress areas (level 1) yield a net marginal effect of $-3.065 + 12.39 = +9.325$ percentage points (interaction significant at 1%), meaning that aid in those areas substantially *increases* conflict probability. Similarly, projects in high water stress areas (level 3) and extreme water stress areas (level 4) produce net conflict-increasing effects of $-3.065 + 4.877 = +1.812$ and $-3.065 + 4.267 = +1.202$ percentage points respectively (significant at 5% and 10%). The interaction term for level 2 is not statistically significant. As a country, Bangladesh faces a moderate overall water stress situation, being placed among the least water-stressed countries globally (Islam and Ali 2019). Nevertheless, it faces specific challenges such as salinity intrusion in coastal areas and seasonal water availability issues during dry seasons (Hedrick 2025), which are qualitatively different from the widespread groundwater over-exploitation that characterises India (UNESCO World Water Assessment Programme 2024). By inspecting the conflict, project location, and water stress maps of Bangladesh (figure 15), we observe that aid projects are widespread across the country with a noticeable concentration in the central and northern parts, while conflicts are concentrated in the western, central, and southern areas. The project title analysis confirms that foreign aid projects in Bangladesh are dominated by infrastructure, education, and governance projects (figure 14), with water-specific projects forming a minority. This sectoral mismatch may explain why aid in water-stressed zones increases rather than reduces conflict: projects that do not directly address the specific nature of Bangladesh’s water vulnerability—seasonal scarcity and salinity—may fail to ease

resource tensions and instead attract competition over newly available external resources.

Nepal's result shares the same directional pattern as Bangladesh for the standalone project location count, with one additional project location decreasing conflict probability by 3.883 percentage points (significant at 5%). However, the only statistically significant interaction term is for medium-to-high water stress areas (level 2), where the interaction coefficient is +4.230 (significant at 1%), yielding a net marginal effect of $-3.883 + 4.230 = +0.347$ percentage points—a near-zero and conflict-increasing result. The interactions for levels 1, 3, and 4 are not statistically significant. In comparison to other countries in the region, Nepal is endowed with abundant water resources sustained by Himalayan meltwater and rainfall-replenished aquifers. Nevertheless, Nepal ranks high in baseline water stress due to widespread contamination of domestic water sources, malfunctioning water supply and irrigation infrastructure, and heightened vulnerability from poverty and natural disasters (Suwal 2025; Udmale et al. 2016). The presence of aid projects overall reduces conflict onset, likely by stabilising infrastructure deficits and improving access to clean water. However, the near-zero net effect in level 2 areas suggests that aid there fails to provide a clear pacifying benefit. In those zones, the overwhelming majority of projects are concentrated in water supply and sanitation to the exclusion of other sectors (see figure A.4 in the separate document *Appendix - continued*), unlike in other water stress level areas where projects are more evenly distributed across transport, governance, and agriculture. The project title analysis confirms this over-concentration (figure 18). Our interpretation is that because Nepal's water deficiency manifests across multiple fronts simultaneously, a narrow sectoral focus in aid allocation may destabilise rather than stabilise affected areas, creating conditions for resource grabs and disputes even when the aggregate aid effect is neutral.

Pakistan presents a distinct result pattern from the other three countries. The standalone project location count coefficient is 3.379, but it is only weakly significant at the 20% level and should be interpreted with caution. The only robustly significant project-related result is the interaction term for low-to-medium water stress areas (level 1), with a coefficient of -6.223 (significant at 5%), yielding a net marginal effect of $3.379 - 6.223 = -2.844$ percentage points. This suggests that well-targeted aid in moderately water-stressed areas can reduce conflict probability, consistent with the finding for India. The interactions for levels 2, 3, and 4 are not statistically significant. A notable result for Pakistan is the GDP coefficient: a 1% increase in GDP raises the conflict onset probability by 47.29 percentage points (significant at 1%), a direction entirely opposite to that observed in India. Under normal circumstances, economic development would be expected to ease tensions by improving living standards; the positive sign here likely reflects the highly unequal distribution of economic gains in Pakistan, where growth has not translated into broad-based improvements. Indeed, Pakistan's Gini index is the worst in the South Asia region at 29.6, compared to 32.4 for Bangladesh and 35.7 for India (Burki 2023), suggesting that GDP growth in Pakistan may concentrate resources among elites, fuelling rather than dampening grievances. Additionally, average precipitation is positive and highly significant (coefficient 0.3479, at 1%), reinforcing the pattern seen in India and Nepal that excessive rainfall increases flood risk and social tensions. Pakistan is ranked by the WRI among countries with high national baseline water stress, facing challenges from outdated water infrastructure, poor governance, and drought risk despite the presence of the Indus River Basin (Sleet 2019). The sector distribution graphs show that in level 1 areas, projects are more evenly distributed across crucial sectors compared to other water stress levels, where there is an overfocus on general government and civil society (see figure A.5 in the separate document *Appendix - continued*). This sectoral balance in level 1 areas likely contributes to their conflict-reducing effect,

and the absence of significant effects at higher stress levels suggests that those more severe and complex problems cannot be resolved by the existing scale of aid allocation alone.

4 Conclusion

To summarize the paper, we aimed to investigate the effect of foreign aid project allocation on the onset of local conflict, particularly under varying water stress conditions in the areas. Using the 50 by 50 km grid cell as the baseline unit of the study and the temporal resolution of 1995 to 2014 and taking into account entity- and time-fixed effects, we applied the grid cell to the country of India. We also checked for the robustness of the result at a finer grid cell resolution of 25 by 25 km and did the external validity for South Asian neighbors.

The primary finding reveals that while foreign aid projects increase conflict probability by 2.983 percentage points per additional location in areas with no water stress (the base category, level 0), this effect is reversed in areas with low-to-medium (category 1) water stress, where the net marginal effect amounts to a reduction of 2.899 percentage points ($2.983 - 5.882$). In extreme water stress areas (category 4), the net marginal effect is near zero ($2.983 - 2.944 = 0.039$), suggesting that aid nearly fully offsets any conflict-inducing tendency in the most water-scarce settings. Furthermore, economic growth, measured by the logarithm of GDP, lowers the conflict onset probability of 0.3881 point per 1 % increase, underscoring the peacebuilding role by improving the living standard for the general population. The extension of the study to the three subcategories of conflict and the logarithm of the best death estimate as dependent variables provides less complete result interpretations compared to the general conflict dummy (Section 3.1.2). Adjusting the study's baseline unit to a 25 by 25 km grid for a robustness check altered the spatial dynamics, resulting in differences in the magnitude of the robustness results compared to the default findings. However, the direction of the effects and the overall conclusions remained consistent between the two analyses. For the literature contribution, this paper contributes to the existing foreign aid–conflict nexus with the unique approach of grid analysis for each individual country and the focus on the water stress to represent a prominent resource constraint in many countries around the globe.

For policy implications of India, aid should be administered in areas with low-to-medium water stress, where resource scarcity is not yet a critical issue, to strengthen community resilience. Alternatively, aid can be distributed to regions facing urgent needs, providing immediate relief to mitigate crises. Furthermore, the local authority could level the benefit of external resources received from foreign aid with its own efforts of developing the local economy. Regionally, while India benefits from targeted aid based on the resource constraint, different result patterns for Afghanistan, Bangladesh, Nepal, and Pakistan (Section 3.3) recommend that administrators should assess carefully the conditions of each country as well as its local communities and allocate the projects accordingly.

Nevertheless, this study is not without default. Firstly, the data of this study is only for the 20-year period from 1995 to 2014, mainly because of the limited temporal resolution of the AidData dataset. And while India is not a big recipient of Chinese aid, the inclusion of Chinese aid to both India and other South Asian countries could improve the power of explanation of the study. And the data extrapolation of some variables due to temporal resolution constraints is an element of contention that the study does

not have the precise data on some variables. The arbitrary choice of 50 by 50 km grid for the baseline unit of study or the 25 by 25 km grid cells of the robustness check could be replaced and tested with a different approach that explains relevant characteristics to the water stress situation. For example, the transboundary nature of water resources across regions as mentioned in subsection 1.2.1 could be somehow included to the base unit of the study. Furthermore, the 25-casualty threshold of the UCDP dataset means that lower-intensity forms of conflict—including resource-related disputes that fall below this threshold—are not captured, which may understate the true relationship between water stress and social tensions. Finally, we recognize that readers should be cautioned to generalize the result pattern of India to other countries because of the limited external validity tests.

At the time of writing this paper (March 2025), the foreign aid sector is undergoing a significant reform. The US government has just launched a comprehensive audit and structural overhaul into the United States Agency for International Development (USAID) (The White House 2025). Discussions about revising foreign aid policies are also happening in other Western countries. With the trend of realignment of foreign aid policies with the interests of donor countries, it could force aid-recipient nations to prioritize their own nation-building efforts and gradually reduce the importance of foreign aids in its economy. Intriguingly, India—being a big recipient of foreign aid—has emerged as a donor country through its own foreign assistance program, providing financial support to neighboring countries in South Asia, as well as nations in Africa and beyond (Ammar Nainar 2024). This aid policy does not only strengthens India’s foreign relations but also serves as a counterweight to Chinese influence in the region. Besides, one certain thing is that, development problems such as water stress, will continue to exist in the future, as highlighted by WRI 2023. While recipient countries can still benefit from foreign assistance, they should focus more on building and improving their own institutions to tackle current and future problems. This dual strategy of taking advantage of external support while making more effort independently on nation building will be essential for achieving sustainable progress and stability in the future.

5 Reference

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6 Appendix

6.1 External validity - 25 by 25 km grid cells

25 by 25 km grid cells					
Dependent Variable: Conflict Dummy					
Country:	Afghanistan	Bangladesh	Nepal	Pakistan	Sri Lanka
<i>Variables</i>					
Project Location Count	-4.705** (2.052)	-0.1142 (0.8602)	-7.517*** (0.9957)	-3.671** (1.635)	-27.58* (16.03)
Project Location Count $\times$ Water Stress Level 1	3.394 (3.231)	2.480* (1.312)	0.6588 (1.887)	7.376* (4.159)	
Project Location Count $\times$ Water Stress Level 2	2.829 (3.225)	-0.0576 (1.275)	0.1427 (2.836)	0.0233 (2.670)	22.74 (17.52)
Project Location Count $\times$ Water Stress Level 3	7.681 (5.570)	0.9759 (1.202)	1.935 (2.131)	4.583** (2.227)	
Project Location Count $\times$ Water Stress Level 4	6.207** (2.881)	0.5056 (1.003)	2.982 (1.812)	4.557** (2.015)	
Log GDP	-0.0904 (11.86)	-27.66* (14.39)	-18.05* (9.301)	35.49*** (6.706)	530.8 (624.0)
Log Population	29.56** (14.62)	-33.38 (21.41)	1.538 (3.681)	6.292 (8.123)	31.26 (38.01)
Average Temperature	6.742 (5.082)	-13.08 (17.39)	1.635 (10.55)	1.342 (1.188)	-208.0 (1,025.6)
Average Precipitation	0.0912 (0.1687)	-0.0402 (0.0525)	0.1283** (0.0622)	0.1114*** (0.0366)	-0.4933 (1.136)
Cropland Percentage	325.2*** (51.16)	-15.46 (22.13)	-117.5* (66.71)	-0.2630 (11.11)	-147.5 (1,020.9)
<i>Fixed-effects</i>					
Grid Cell	Yes	Yes	Yes	Yes	Yes
Subdistrict * Year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	20,600	4,480	4,740	27,920	2,140
R ²	0.70485	0.49381	0.50051	0.47144	0.95190
Within R ²	0.02204	0.00257	0.02258	0.01059	0.05971

Clustered (*grid_id*) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 6: Robustness for external validity tests of South Asian countries

With the finer grid cell resolution (25 by 25 km), results of other South Asian countries have more statistically significant coefficients for the case of Afghanistan, Pakistan, and Sri Lanka, when compared to the default 50 by 50 km grid cells in subsection 3.3. Except for Bangladesh, the standalone project location count (in level 0 areas) is significant and conflict-reducing across all countries. The effects

are especially outstanding for Nepal and Sri Lanka, with one additional project location decreasing conflict onset probability by 7.517 and 27.58 percentage points, respectively. For Afghanistan, the conflict-reducing effect of the standalone project count (-4.705) is partially offset in high and extreme water stress areas (level 4), where the net marginal effects become positive at $+1.502$ percentage points, suggesting that aid in Afghanistan's most water-constrained zones can heighten rather than ease tensions. The interaction term for level 2 loses its significance for Nepal when compared to the default 50 by 50 km result, suggesting that finding was sensitive to spatial aggregation. Pakistan's result offers the most complete picture: administering projects in low to medium (level 1), high (level 3) and extreme (level 4) water stress areas yields net conflict-increasing effects of $+3.705$, $+0.912$ and $+0.886$ percentage points respectively, outweighing the general conflict-reducing benefit of project allocation in level 0 areas. In essence, the robustness check of the external validity reaffirms the relative trade-off between a coarser study unit with higher overall explanatory power (R^2) and the more granular grid cell resolution, which unveils more localised and context-dependent dynamics.

6.2 Maps and graphs of other South Asian countries

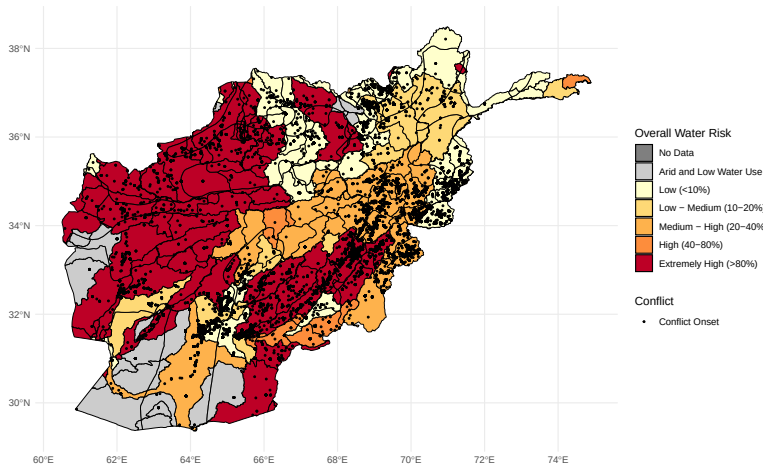


Figure 8: Water Stress and Conflict Locations in Afghanistan

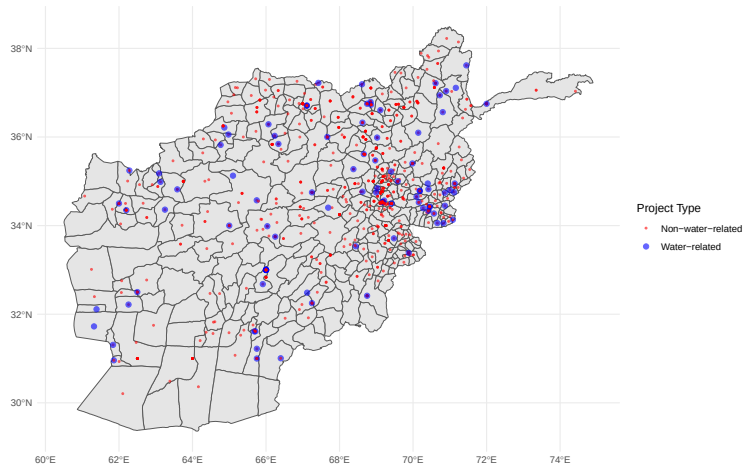


Figure 9: World Bank Aid Projects in Afghanistan from 1995 to 2014

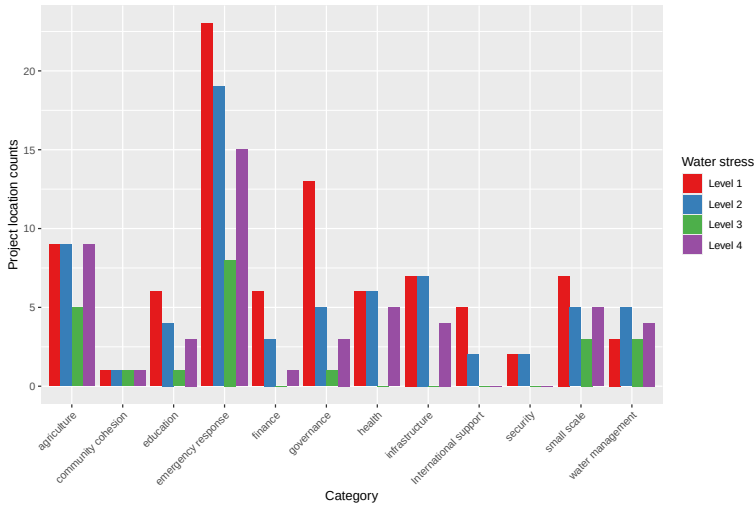


Figure 10: Project Title Analysis for Afghanistan
Note: one project could be categorized in more than one category and could have more than one location

Figure 11: Overview of Water Stress, Conflict, and World Bank Projects in Afghanistan

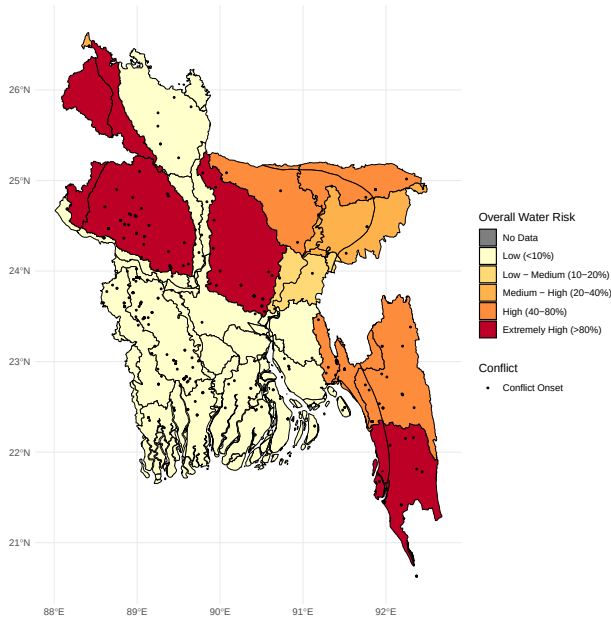


Figure 12: Water Stress and Conflict Locations in Bangladesh

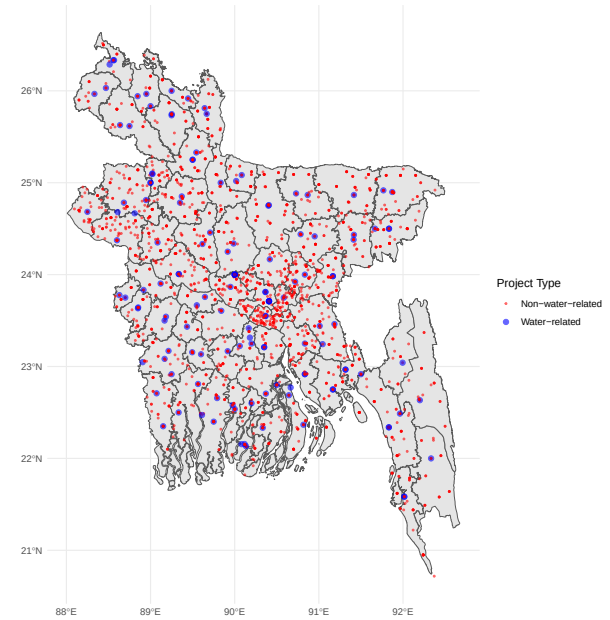


Figure 13: World Bank Aid Projects in Bangladesh from 1995 to 2014

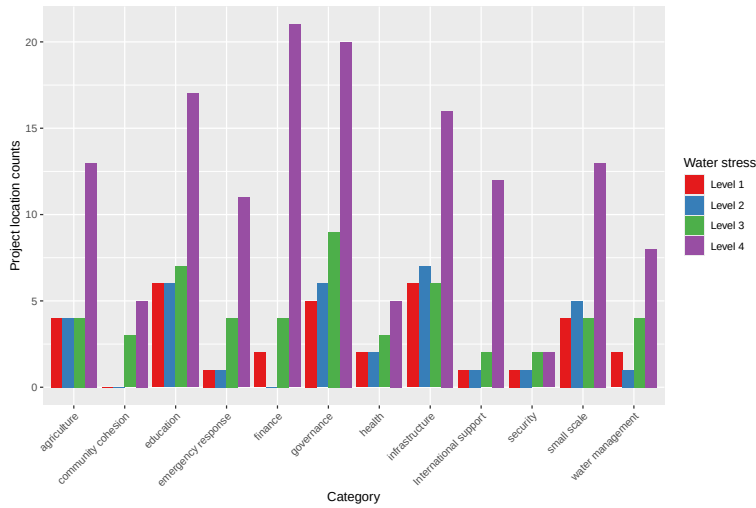


Figure 14: Project Title Analysis for Bangladesh
Note: one project could be categorized in more than one category and could have more than one location

Figure 15: Overview of Water Stress, Conflict, and World Bank Projects in Bangladesh

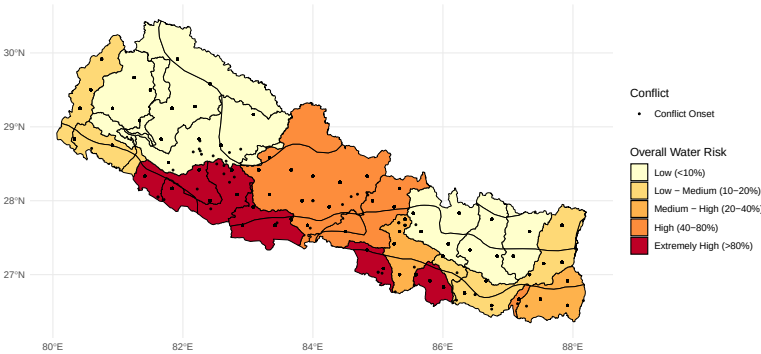


Figure 16: Water Stress and Conflict Locations in Nepal

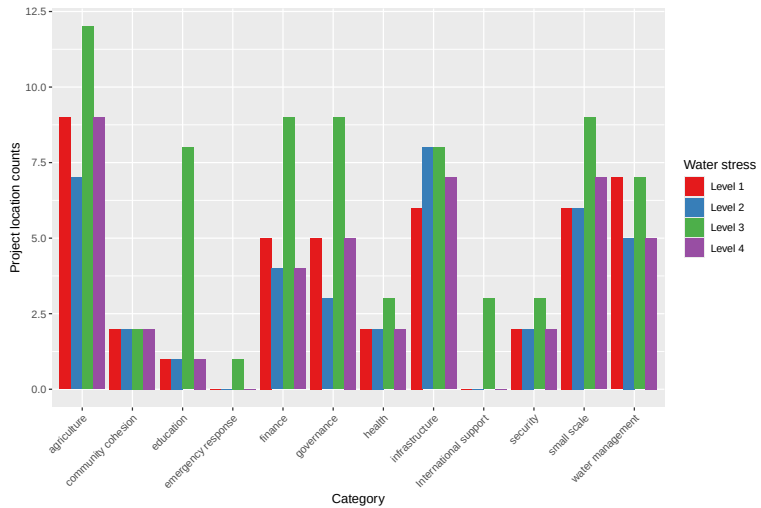


Figure 18: Project Title Analysis for Nepal

Note: one project could be categorized in more than one category and could have more than one location

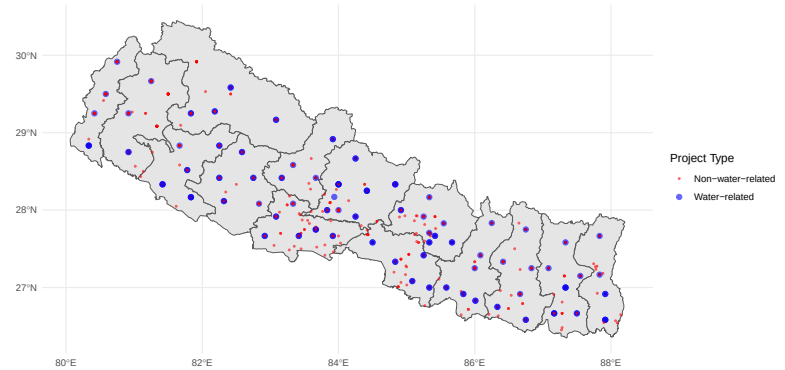


Figure 17: World Bank Aid Projects in Nepal from 1995 to 2014

Figure 19: Overview of Water Stress, Conflict, and World Bank Projects in Nepal

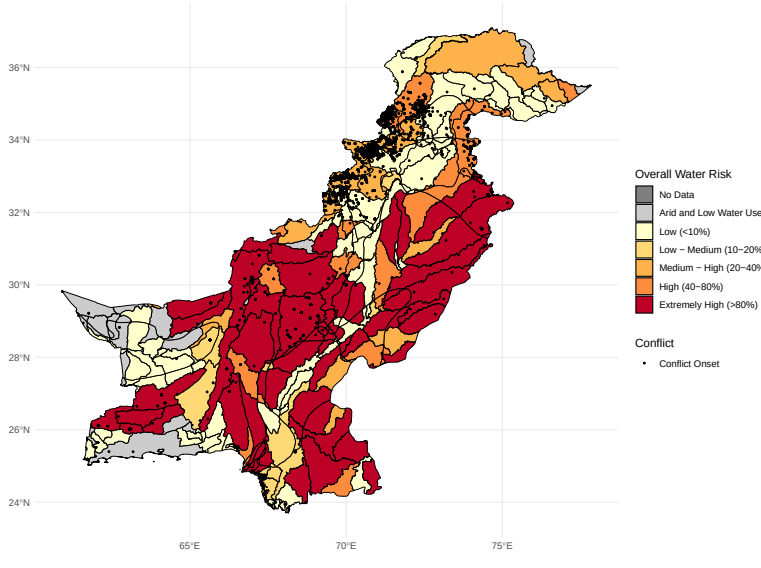


Figure 20: Water Stress and Conflict Locations in Pakistan

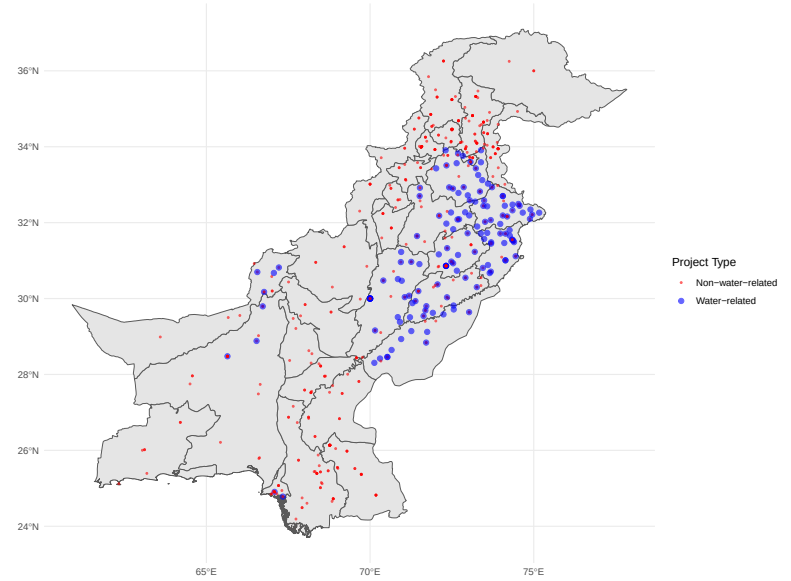


Figure 21: World Bank Aid Projects in Pakistan from 1995 to 2014

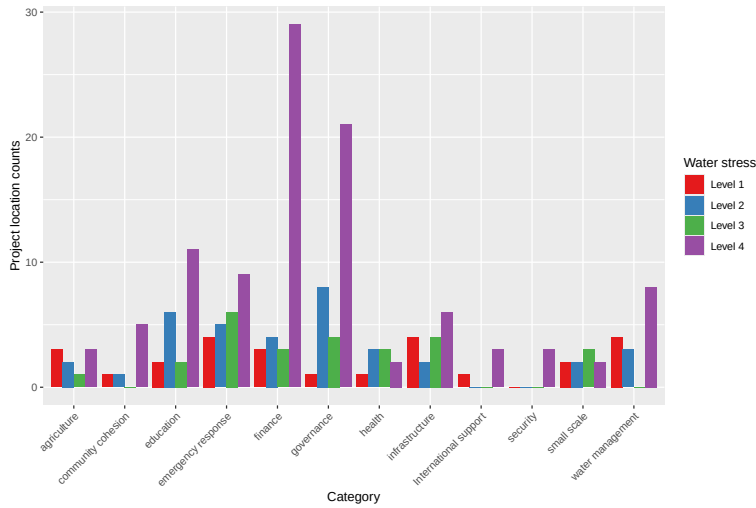


Figure 22: Project Title Analysis for Pakistan
Note: one project could be categorized in more than one category and could have more than one location

Figure 23: Overview of Water Stress, Conflict, and World Bank Projects in Pakistan

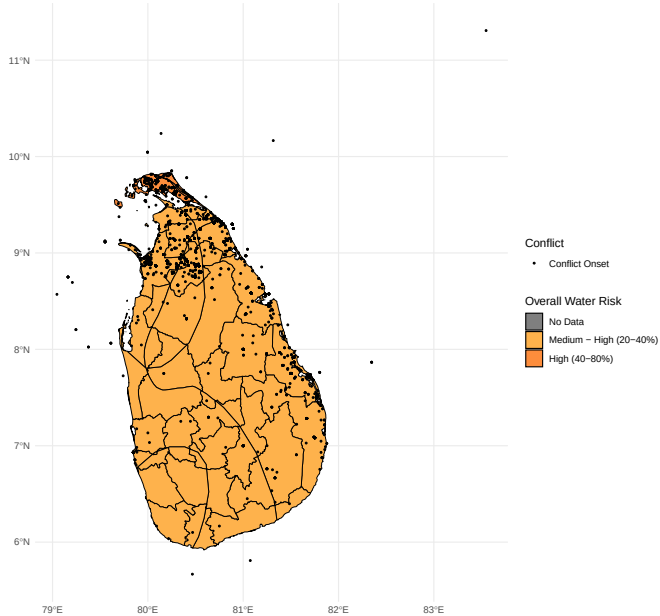


Figure 24: Water Stress and Conflict Locations in Sri Lanka

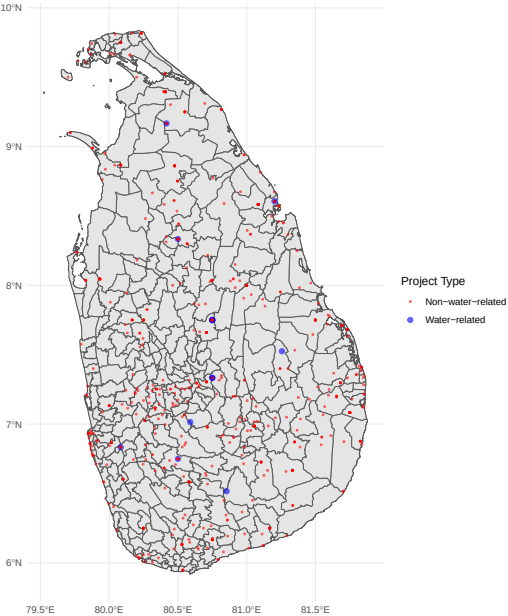


Figure 25: World Bank Aid Projects in Sri Lanka from 1995 to 2014

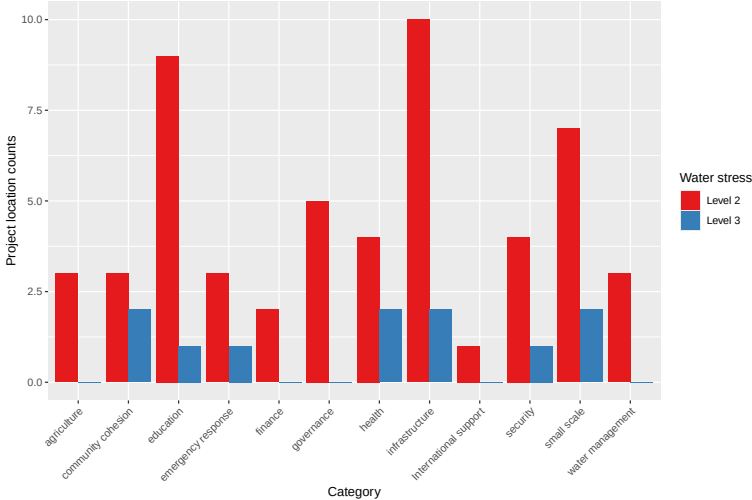


Figure 26: Project Title Analysis for Sri Lanka
Note: one project could be categorized in more than one category and could have more than one location

Figure 27: Overview of Water Stress, Conflict, and World Bank Projects in Sri Lanka